Target-Based Communication under Credibility-Distorted Beliefs

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ABSTRACT: Angeletos and Sastry (2021) theoretically show that central bank (CB) target-based communication anchors inflation expectations better than its interest rate counterpart. However, more UK households report adjusting their expectations based on the interest rate, they under-react to new information, and experienced inflation lowers credibility among the cohorts that have lived through the strongest inflationary episodes. This raises the question of whether intensively investing in target-based communication would anchor expectations and deliver welfare improvements in this context. I explore it by proposing an expectation mechanism that unifies experience-based expectations with endogenous credibility, distorted by the diagnostic expectations of Bordalo et al. (2018), and incorporate it into a Representative Agent (RANK) and a Heterogeneous Agent (HANK) New Keynesian model. Next, I characterize the welfare cost of distorted beliefs as the sum of a diagnostic error, common to every cohort and beyond the reach of communication, and a credibility error, inherent to cohorts that experienced high inflation, which I argue is the only part the CB can influence. The main findings are: i) in both models the loss is carried by the cohorts who remember differently, and it is smaller in the HANK model because constrained households substitute little over time; ii) communicating the target lowers the dispersion of beliefs but raises the error of the consensus, so the optimal intensity is higher in the HANK model, where the older cohorts over-estimate inflation for longer; and iii) in the RANK model communication adds no additional gain to an aggressive interest rate rule, whereas in the HANK model the two are complements and communicating on impact is better. Therefore, target-based communication is a useful and complementary tool for a CB when credibility is eroded.

1. Introduction

When inflation returned in 2021–2023, central banks, CB, henceforth, around the globe asked households to trust that it would come back to the target rapidly. In 2022, when inflation reached four-decade highs, central bankers spoke with one voice, confirming their commitment to return inflation to the target in an attempt to prevent a de-anchoring of expectations. For example, at the 2022 Jackson Hole symposium, the Federal Reserve Chair, Jerome Powell, stated that the "overarching focus right now is to bring inflation back down to our 2 percent goal" and that "we will keep at it until we are confident the job is done" (Powell 2022), a message he repeated a year later when he reminded the public that "two percent is and will remain our inflation target" (Powell 2023). Another example is the European Central Bank, which made the same request in November 2022, when its President, Christine Lagarde, argued that "the more confident the public is that inflation will return to 2% in a timely manner, the smoother the adjustment process will be" (Lagarde 2022), or the Bank of England, which in the same November told households that "it's our job to make sure that inflation returns to our 2% target" and that it expected "inflation to fall sharply from the middle of next year" (Bank of England 2022).

Communicating the inflation target has become a recurrent strategy of CB, with both theoretical and empirical support. On the one hand, Angeletos and Sastry (2021) theoretically show that communicating the inflation target anchors expectations better than communicating the policy instrument, while, on the other hand, Coibion et al. (2022) empirically find that communicating a simple statement about the 2% inflation target lowers inflation expectations as much as what the full policy statement does, and D'Acunto et al. (2020) find that target-based communication moves household expectations more than what an interest rate statement does as well.

However, for UK households these results contrast with what they themselves report when asked about the reasons behind the change in their inflation expectations. Available data until 2026 from the Bank of England Inflation Attitudes Survey show that, on average, 26% of them cite the level of interest rates, whereas only 16% cite the inflation target, and this gap is consistent across age, income, and education, among other characteristics. This raises the policy question of the effects of using target-centred communication to reach a larger share of individuals,

given its documented advantages in anchoring inflation expectations.

This paper analyse this question through the lens of distorted beliefs that incorporate how credibility is shaped by inflation experiences under diagnostic expectations, DE henceforth, two additional empirical regularities for UK households documented in this paper. The data show an under-reaction when new information is incorporated into inflation forecasts, in line with the consensus-level evidence of Bordalo et al. (2020) but with the opposite sign to the over-reaction that diagnostic expectations were designed to capture. Likewise, households that have experienced a higher inflation rate during their lifetime tend to display a lower level of credibility, in line with Salle et al. (2023), who find that remembered price surges lower trust in the CB.

To rationalize these empirical facts, I propose a mechanism that unifies experience-based expectations with endogenous credibility in how agents form the subjective distribution of inflation from which they then retrieve their point expectations. Intuitively, agents are uncertain about the anchor around which inflation gravitates, which is either the inflation they have experienced or the inflation target. This is precisely where CB communication is introduced, since it intentionally pulls the experienced anchor toward the inflation target. Through Bayesian updating, I define the subjective probability that the agent assigns to the uncertain anchor being the inflation target as subjective credibility, which is then endogenously incorporated into the agent's predictive distribution of inflation.

Intuitively, distorted beliefs determine different ex-ante real interest rates across age cohorts, which, for the same inflationary shock, cause different consumption responses between cohorts. Since every cohort under-reacts to the shock, it expects too little inflation, perceives an ex-ante real rate that is too high and non-optimally postpones consumption relative to the path it would choose with correct beliefs. For the older cohorts, with a higher inflation experience and lower credibility, the forecast leans on lived inflation, so after impact they expect higher inflation, perceive a lower ex-ante real rate and non-optimally bring consumption forward. The non-optimal consumption responses relative to the path with correct beliefs, the consumption wedges henceforth, derived from the different perceived rates naturally constitute the measure on which I analyze the welfare implications of distorted beliefs.

By embedding the distorted beliefs mechanism into a general equilibrium framework such as a Representative Agent New Keynesian (RANK) model and a

lifecycle Heterogeneous Agent New Keynesian (HANK) model, next I study the welfare effects of the intensity with which the central bank communicates its target. I show that the consumption wedges are the sum of two forecast errors. First, the *diagnostic error*, which captures the households' under-reaction, is negative for every cohort and beyond the reach of communication. Second, the *credibility error*, which follows from the households' inclination towards lived inflation, is positive for the older cohorts whose inflation experience diverges consistently from the target, and is the only part the CB can move.

The findings are summarized as follows. First, in both models the welfare loss from beliefs is concentrated on those who remember differently, but this effect is weaker in the HANK model due to the real-rate and earnings-heterogeneity channels. Importantly, the younger cohorts, with the highest Marginal Propensity to Consume (MPC) and the smallest asset positions, cut consumption the most after the shock, but this is the incidence of the shock and not a loss from beliefs. However, even if beliefs are a small part of the consumption response, they remain the only part that the CB can reach with communication.

The second result is concerned with the superiority of target communication discussed in Angeletos and Sastry (2021) and Coibion et al. (2022). Communicating the target removes the credibility error of every cohort, but it is not unequivocally superior. To show this, the cohort welfare measure I propose is the norm of its consumption wedges over the whole horizon of the shock, and the aggregate measure weights these norms by population shares. This measure has the appeal of decomposing into the dispersion of the wedges across cohorts and the error of the consensus, which allows a cost-benefit analysis of communication. The benefit is that communication lowers the dispersion, particularly for the older cohorts with a higher inflation experience, which would have forecast higher inflation after the shock but are anchored to the target by communication. The cost, on the other hand, is captured by the increase in the error of the consensus, given that those forecasts were the only part of the consensus that followed the realised inflation.

The optimal intensity balances these two effects showing that the result is determined by the different behavior of the older cohorts across models. In RANK, allowing older cohorts to incline towards their inflation experience, which requires little target communication, reduces the cost of communication measured by the consensus error. In the HANK model this result changes because of the nature

of wealth heterogeneity. After an inflationary shock, the high-MPC households close to the budget constraint cannot smooth the fall in labour income, so the contraction in demand is stronger and more persistent than in the RANK model. Inflation therefore returns to the target faster, but the older cohorts keep over-estimating it for longer given their high inflation experience. Hence, the main benefit of communication is that it stops this over-estimation of the older cohorts.

The third result connects target communication with traditional monetary policy by analysing their interplay. For the RANK model, a more aggressive interest rate response reduces the dispersion and the consensus error in the same proportion, whereas intensive target communication raises the consensus error by as much as it removes the credibility error. Hence, there is no substantial gain from using both target communication and aggressive monetary policy. For the HANK model, the results are the opposite, because the aggressive response pushes inflation below the target sooner and the older cohorts keep over-estimating it, and intense communication removes precisely this over-prediction. Consequently, intense target communication and a moderately aggressive rule deliver the lowest loss, and the two policies become complements. Finally, the timing of communication matters: in the RANK model a short delay is better than communicating on impact, since it allow the older cohorts' experience pull the consensus toward realised inflation for a few periods, whereas in the HANK model communicating on impact is best, because it prevents credibility from eroding while the over-prediction episode is still in place.

This paper is connected with the the literature as follows. First, with the literate on measuring CB credibility. In his seminal article, Blinder (2000) settles on the definition that a CB is credible if people believe it will do what it says. There are, at least, two groups that have empirically tried to quantify CB credibility. The first, following a reduced-form approach, measures it from the distance between expectations and the CB inflation target, as in Bomfim and Rudebusch (2000) and Bordo and Siklos (2015), or by asking people directly, as in Haldane and McMahon (2018).

On the other hand, the second group makes credibility endogenous, dependent on past errors. For instance, Hommes and Lustenhouwer (2019) model it as the fraction of the population that trusts the target rather than extrapolating past inflation, consistent with McMahon and Rholes (2025), who show experimentally that people judge the central bank by its past forecast errors. In parallel, bringing

in a long-run experiential component, Malmendier and Nagel (2016) show that households form inflation expectations from the inflation they have personally lived through, and Salle et al. (2023) find that remembered price surges lower trust in the CB. This paper connects both strands by modelling credibility as a belief that each cohort updates against its own inflation history, and by introducing it as a probability inside the agent's mind, which uses the same functional form as the population fraction of Hommes and Lustenhouwer (2019), delivering a predictive probability distribution with endogenous credibility.

The second literature branch that this paper relates studies how inflation expectations respond to news, particularly, whether they over- or under-weight them in their predictions. In an influential paper, Coibion and Gorodnichenko (2015) document that consensus forecasts under-react to news, and Bordalo et al. (2020) have shown that under-reaction at the consensus level may coexist with over-reaction at the individual level. Using the diagnostic expectations (DE) of Bordalo et al. (2018), it is possible to reconcile these over- and under-reaction differences. Simultaneously, L'Huillier et al. (2024) and Bianchi et al. (2024) have incorporated DE into the New Keynesian model, which helps to analyse general equilibrium questions about this sort of distortion. This paper borrows the DE operator and applies it to the predictive distribution of inflation expanded by credibility, which allow me to distorting inflation expectations.

The third research area to which this paper connects is CB communication with the public. There is ample evidence that communication works well to anchor the expectations of the most informed agents in the economy, such as professional forecasters and financial markets. For example, Romer and Romer (2000) show that private forecasters update taking as reference the new information the Fed reveals through its policy actions, or Nakamura and Steinsson (2018) who show that policy announcements move professional forecasts and asset prices at high frequency. However, communicating with the general public has been a gigantic challenge for the CB and has often been unsuccessful, as analysed by Binder (2017) and Lamla and Vinogradov (2019), who find that policy announcements are largely unnoticed by households, and by Kumar et al. (2015), who show that even after decades of inflation targeting, firms' inflation expectations in New Zealand remain essentially unanchored. This lack of reach of the CB to the public limits the beneficial results of target communication documented by Angeletos and Sastry (2021), D'Acunto et al. (2020) and Coibion et al. (2022) discussed above. This paper

takes that evidence and seeks to contribute to the discussion on how investing heavily in communicating the target can significantly improve household welfare.

Finally, this paper is also related to the literature on heterogeneous beliefs and redistribution in New Keynesian models. Auclert (2019) shows that the covariance between MPCs and URE determines the winners and losers of monetary policy. Guerreiro (2023) and Shabalina and Tzaawa-Krenzler (2024) independently study how imperfect information interacts with wealth heterogeneity by adding an additional covariance, between MPCs and beliefs, which come from disagreement or attention, respectively. This paper borrows the cohort belief structure of Guerreiro (2023) and the sequence-space representation of Auclert et al. (2021) to solve HANK models to analyse who gains, and how much, from CB communication when expectations are distorted in the way I propose. The rest of this paper is organized as follows. Section 2 presents the stylized facts, Section 3 develops the inflation expectation formation process, Section 4 embeds it into the RANK and HANK models, Section 5 presents the results on target communication and welfare, and Section 6 concludes.

2. Empirical Motivation

This section presents the stylized facts about inflation expectations for the UK that motivate this paper. Throughout this section, a mix different data sources are used. I make intensive use of the Bank of England Inflation Attitudes Survey, the source for inflation expectations and for the preferred measure of central bank credibility. The cohort inflation histories are built from the Consumer Price Index published by the ONS, and the wealth measure comes from the ONS Wealth and Assets Survey. A detailed explanation of the variable construction and may be found in Appendix C, while subsection C.5.1 is reserved for descriptive statistics.

2.1. Stylized Facts

Fact 1. UK households under-react to inflation news.

"Household inflation expectations under-reaction to news."

The evidence comes from applying into our sample the forecast-error-on-revision test of Coibion and Gorodnichenko (2015), which evaluates whether expectation revisions predict the subsequent forecast errors. If this is true, then,

the forecasts have not fully incorporated the news, which is evidence of a degree of information rigidity. For US professional forecasters in the period 1969–2014, they find a coefficient around 1.2 for the inflation rate, and the same positive sign for US consumers and firms, and for professional forecasters in twelve advanced economies, the UK among them.

In this spirit, Bordalo et al. (2020) apply an analogous test to an ample set of US series from the SPF and Blue Chip surveys over 1968–2016 and find that over- or under-reaction depends on the level of aggregation. On the one hand, at the aggregate level (consensus) there is under-reaction[1] as in Coibion and Gorodnichenko (2015), but at the individual level, forecasters over-react to their own revisions. They reconcile both facts with diagnostic expectations (Bordalo et al. 2018) under dispersed information, and estimate a diagnosticity parameter of about 0.5, which is in line with the values that L'Huillier et al. (2024) and Bianchi et al. (2024) independently estimate when incorporating Diagnostic Expectations (DE) into the New Keynesian model.

I visually implement the test in Equation 1 using the BoE Inflation Attitudes Survey, with quarterly data from 2010Q1–2024Q4, with the caveat that its design is a repeated cross-section, thus, only the consensus version of the test may be performed:

$$\underbrace{\pi_{t,t+3} - \mathbb{E}_{g,t}[\pi_{t,t+3}]}_{\mathrm{FE}_{g,t}} = \beta_{0,g} + \beta_{1,g} \underbrace{\left(\mathbb{E}_{g,t}[\pi_{t,t+3}] - \mathbb{E}_{g,t-4}[\pi_{t,t+3}]\right)}_{\mathrm{Rev}_{g,t}} + e_{g,t}. \quad (1)$$

Next, I provide a brief explanation of the elements of this regression. First, $\pi_{t,t+3}$ is the realised CPI inflation over quarters t to $t+3$, and is measured as the average of the four quarterly annualised rates. Second, $\mathbb{E}_{g,t}[\pi_{t,t+3}]$ is the cohort's consensus expectation over the same quarters, and $\mathbb{E}_{g,t-4}[\pi_{t,t+3}]$ is the expectation the same cohort held one year earlier for the same twelve months.[2]

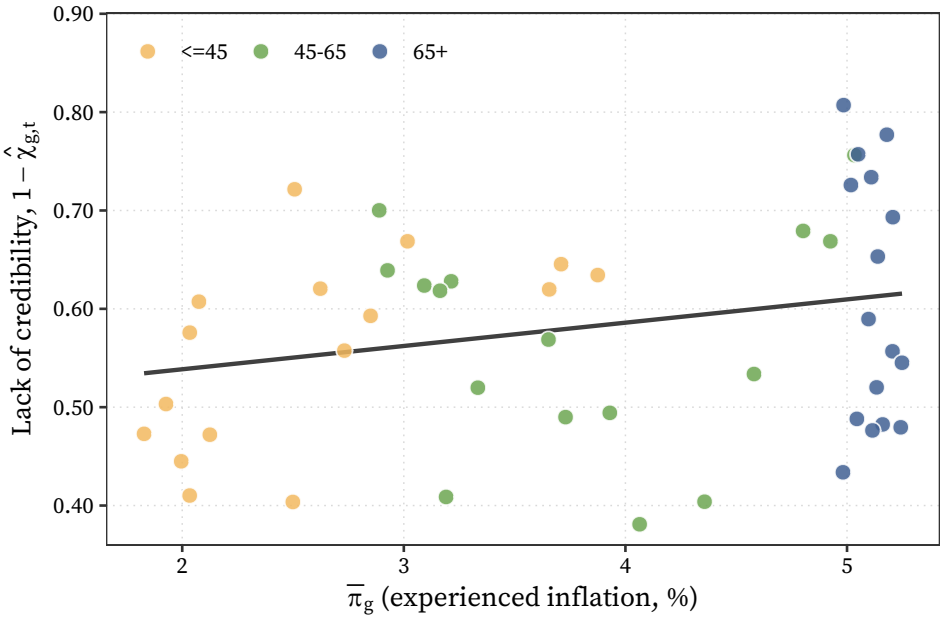
In the spirit of Bordalo et al. (2022), revisions and errors are calculated on a fixed-event basis to keep a consistent comparison window. The expectation revisions, $\mathrm{Rev}_{g,t}$, show how expectations were updated today compared with one

[1] Similar under-reaction to macroeconomic news, including inflation, is documented for Italian and German firms by Born et al. (2025).

[2] The different expectation's horizon, I use Q2a that asks "How much would you expect prices in the shops generally to change over the next 12 months?" and Q2b, that follows with "And how about prices the year after next? How much would you expect prices in the shops generally to change between [month, year +2] and [month, year +3]?".

year ago for the same target window. On the other hand, forecast errors, $FE_{g,t}$, show how close expectations were to realised inflation over the same window. Under-reaction is found when news that raise revisions are accompanied by positive forecast errors, which ultimately implies that the forecast fell short of fully incorporating the news.

FIGURE 1. **Forecast Errors and Revisions**



Note: Sample: BoE Inflation Attitudes Survey, quarterly waves 2010Q1–2024Q4, 60 waves per cohort.

Figure 1 shows a consistent positive slope regardless of age, meaning the three cohorts under-react by a similar amount. Thus, when a cohort revises its inflation expectation upward, realised inflation turns out to be above the revised forecast. Although the revision is made in the right direction, its magnitude is lower than what a full-information response would have implied, providing evidence that only a fraction of the news was incorporated in the revised forecast.

Fact 2. Lived Inflation, Credibility and Inflation Expectations.

"Cohorts that lived through higher inflation are less confident that inflation will return to the target, and cohorts that are less confident expect higher inflation."

This section documents that cohorts that have experienced higher inflation report lower credibility and also expect a higher inflation rate. To provide evidence

on this fact, the credibility measure I use is question Q18 of the BoE Inflation Attitudes Survey with sample 2011Q1–2025Q3, when Q18 is asked. This question asks "How confident are you that inflation will be close to the 2% target set by the government in two to three years' time?"[3] I define the lack of credibility of cohort g in wave t as:

$$1 - \hat{\chi}_{g,t} \equiv \frac{\sum_{j \in \mathcal{V}_{g,t}} w_j \mathbf{1}\{\text{Q18}_j \in \{\text{not very confident},\ \text{not at all confident}\}\}}{\sum_{j \in \mathcal{V}_{g,t}} w_j}, \quad (2)$$

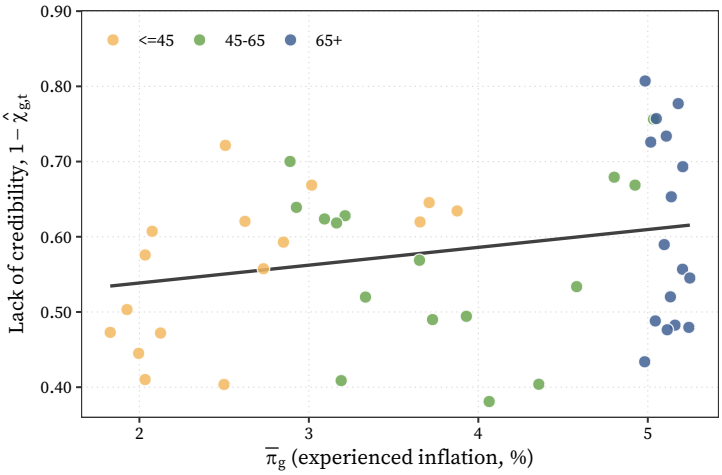
and, similarly, $\hat{\chi}_{g,t}$ is the share of the cohort that answers "very confident" or "fairy" confident.[4] Therefore, we can consider $\hat{\chi}_{g,t}$ as the share of credibility for cohort g. It is closely related to the credibility measure of Bomfim and Rudebusch (2000), in the sense of "the fraction of the population that believes the target will be achieved", and similarly to the fraction of fundamentalists of Hommes and Lustenhouwer (2019), who believe that inflation will be at the target.

Evidence is provided by Figure 2, which plots the cohort proportion self-reported lack of credibility, $1 - \hat{\chi}_{g,t}$, against the experienced inflation $\bar{\pi}_g$. It shows a positive relationship, which supports the hypothesis. This is consistent with the track-record view of Blinder (2000), in which credibility is earned by a history of low inflation, and with Bomfim and Rudebusch (2000), where the weight on the target falls with the past deviation of inflation from it and with Salle et al. (2023), who find that households who remember more price surges report lower trust in the central bank.

[3] By close it means between 1% and 3%. The potential answers are "very confident", defined in the questionnaire as a better than 50% chance, "fairly confident", around 50%, "not very confident", around 25%, and "not at all confident", less than 25%.

[4] The two shares are complements among valid answers, so either one defines the same $\hat{\chi}_{g,t}$.

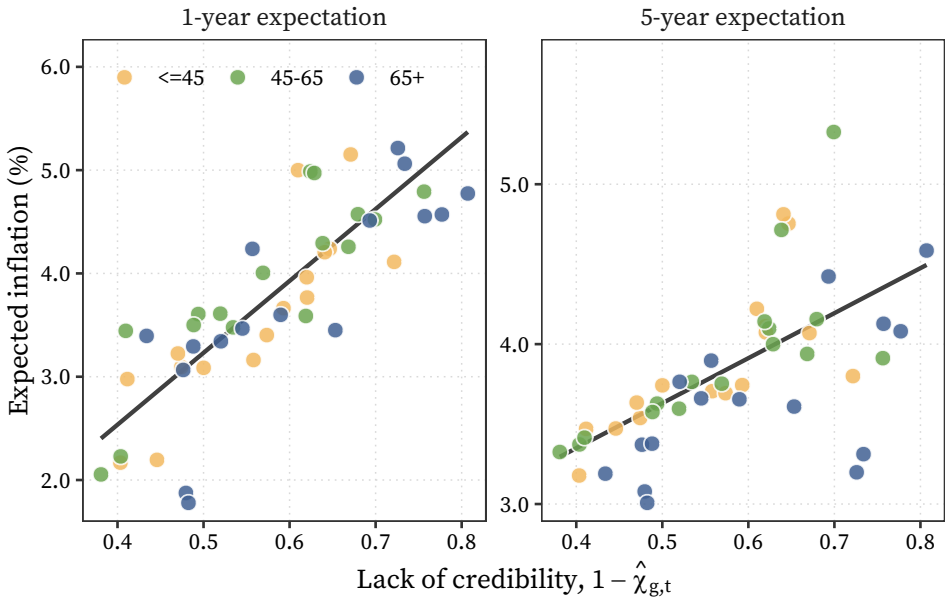
FIGURE 2. **Credibility and Experienced Inflation**



Note: Sample: BoE Inflation Attitudes Survey, waves with Q18, 2011Q1–2025Q3, 16 waves. "don't know" excluded. OLS estimates for the credibility measure for the possible alternatives: i) very confident −0.0067 (−2.39); ii) fairly confident −0.0170 (−3.21); ii) very or fairly ($\hat{\chi}_{g,t}$) −0.0237 (−4.08); iv) not very confident 0.0017 (0.67); v) not at all confident 0.0220 (4.12); vi) **Measure used in the paper:** not very or not at all ($1-\hat{\chi}_{g,t}$) 0.0237 (4.08).

Figure 3 closes the reasoning by plotting the cohort's inflation expectations versus the self-reported credibility. At the two horizons, the slope is positive, meaning that a cohort with a higher proportion of people distrusting that inflation will return to the target expects more inflation. This is consistent with Christelis et al. (2020), who find that households with more trust in the central bank report lower and less dispersed inflation expectations, and with Haldane and McMahon (2018), who show, using the same survey that respondents more satisfied with the Bank of England performance consistently get expectations closer to the inflation target. Conclusively, the two figures say that lived inflation increases the lack of credibility which, in turn, raises inflation expectations. This reasoning motivates the proposed inflation expectations mechanism of Section 3.

FIGURE 3. **Inflation Expectations and Credibility**



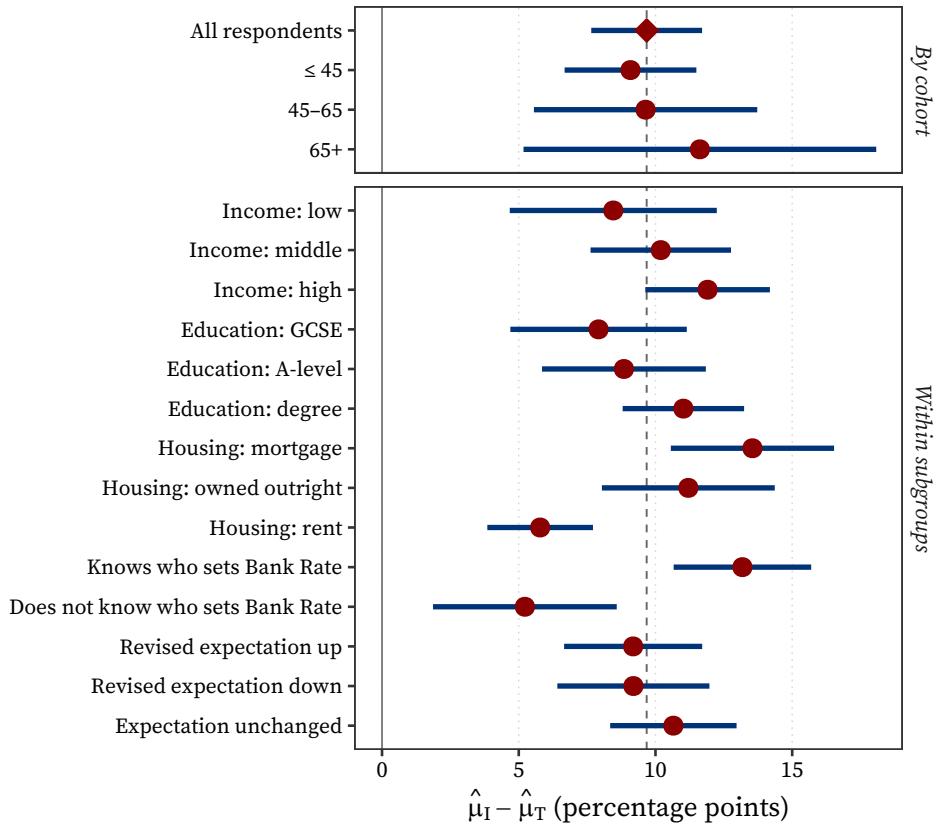
Note: Sample: BoE Inflation Attitudes Survey, waves with Q18, 2011Q1–2025Q3, 16 waves. "don't know" excluded. Slope coefficients: i) one-year horizon 6.96 (5.91); ii) five-year horizon 2.82 (4.49).

Fact 3. Instrument-Over-Target Salience.

"More households report forming their inflation forecast on the interest rate than on the inflation target."

To state this fact, I use question Q2aiv in the survey, which asks households who changed their expectation "What were the most important factors that led you to your expectation of prices in the shops over the next 12 months?". Across the potential reasons, I focus on the cues the central bank controls through its communication: the inflation target, $\hat{\mu}_T$, and the policy instrument, $\hat{\mu}_I$. For each case, I report the mean across cohorts in Figure 4, which shows that the weighted share citing the instrument is statistically higher than the weighted share citing the target in every cohort and within subgroups of the population. This empirical result contrast with what Angeletos and Sastry (2021) theoretically recommend. In their framework, communication about the target anchors expectations more effectively than communication about the instrument when general-equilibrium feedbacks are strong. However, UK households report that they intensively form their inflation expectations around the policy rate.

FIGURE 4. **Instrument Salience: Interest Rate versus Inflation Target**



Note: Sample: BoE Inflation Attitudes Survey, waves with Q2aiv (2017Q1, 2023Q1, 2024Q1, 2025Q1, 2025Q3, 2026Q1). Income rows use the post-2022 sample. Dots are the estimates $\hat{\mu}_I - \hat{\mu}_T$, the vertical line marks the null $\mu_I - \mu_T = 0$, and horizontal lines are the 95% confidence intervals with clustered standard errors.

Three features of Figure 4 matter for the model presented in Section 3. First, the gap is positive and significant for every group, so a higher proportion of households report the interest rate as the cue they consider when changing their expectations. Second, the gap is indistinguishable across cohorts, so the salience of the instrument over the target is a common feature of the whole population rather than a characteristic of the cohorts that lived through high inflation. Finally, the difference is higher for high-income households and for those with exposure to monetary policy fluctuations, such as those with a mortgage and those who know who sets Bank Rate. This result is expected, since households with mortgages are exposed to the policy rate through their mortgage payments and savings, that is, through their balance sheet. This is the unhedged interest rate exposure channel documented by Auclert (2019).

This pattern is consistent with the evidence that central bank communication about its objectives is mainly ignored by the public. Binder (2017) and Lamla and Vinogradov (2019) find that households are mostly unaware of policy announcements, to the point that Haldane and McMahon (2018) describe the general public as the blind spot of central bank communication. Despite this unawareness of the public, there is evidence that the central bank can influence expectations with active target communication. Ehrmann et al. (2025) document that the announcement of the ECB's revised inflation target went largely unnoticed, whereas the same content delivered directly through information treatments raised the credibility of price stability. This is justifies why does the model of Section 3 considers the target as a feature the central bank can choose to actively communicate through the intensity ν.

Taken together, the three facts describe characteristic aspects of how UK households form inflation expectations. They under-react to news (Fact 1); the credibility they have is shaped by the inflation they have lived through, and inflation expectations are affected by it (Fact 2) and; the inflation target is cited less than the interest rate as a cue when changing their expectations (Fact 3). Section 3 takes these facts and formalizes a theory in which agents are uncertain about the anchor around which the inflation rate gravitates, credibility is microfounded within the agent's mind and, finally, under-reaction is an essential component explaining how agents incorporate news and ultimately delivers a distorted beliefs operator that will be incorporated in the upcoming sections.

Two additional facts, documented in Appendix D, show interesting evidence on the potential effects that distorted beliefs may have on the economy. First, households that expect more inflation report bringing forward purchases and saving more (Fact A1). Second, cohorts that lived through higher inflation perceive higher inflation today and have become less confident that higher interest rates slow down price rises (Fact A2).

3. Credibility and Diagnostic Expectations

This section formally introduces how the credibility channel connects with the endogenous-credibility approach of Hommes and Lustenhouwer (2019) and the diagnostic-expectations (DE) framework of Bordalo et al. (2020) and L'Huillier et al. (2024). The connection is not direct, since DE can be used only when a well-defined

subjective distribution is available. Particularly, DE are defined as a distortion of an agent's conditional distribution relative to that same agent's lagged conditional distribution. Therefore, to apply the diagnostic operator we need a forecaster whose beliefs at t and $t-1$ are linked by an updating rule. On the other hand, the endogenous credibility of Hommes and Lustenhouwer (2019) only provides the aggregate proportion of the population that is fundamentalist, its preferred measure of the anchored population and therefore of credibility. However, it leaves unanswered how agents incorporate credibility into their subjective distribution of future inflation at the individual level. Therefore, this section proposes an approach delivering a well-defined distribution for future inflation on which the diagnostic operator can act.

Suppose cohorts indexed by g, each with a representative agent within, as Assumption 1 states.

ASSUMPTION 1 (One forecaster per cohort). *Each cohort g is a single representative forecaster. Within a cohort there is no heterogeneity of beliefs and no averaging across people. Every weight that appears below is a probability held by this one agent, not a share of a population.*

This agent believes that the inflation rate is an AR(1) process around an anchor, m, whose true value is unknown. The perceived law of future inflation is summarized by Assumption 2:

ASSUMPTION 2 (Perceived Law of Future Inflation). *Cohort g believes inflation follows a first-order autoregression around an anchor m,*

$$\pi_{t+1} = \rho_g \pi_t + (1-\rho_g)\, m + \varepsilon_{t+1}, \qquad \varepsilon_{t+1} \sim N(0, \sigma_g^2) \text{ i.i.d.} \tag{3}$$

It knows ρ_g and σ_g, but it does not know m, the level inflation returns to.

Each agent balances two hypotheses about the anchor. First, under hypothesis T, the anchor is the central bank's inflation target. Second, under hypothesis E, the anchor is a communication-weighted sum of the inflation target and the lived inflation, $\bar{\pi}_g$. Introducing $\bar{\pi}_g$ as a potential anchor has ample empirical support. For instance, Malmendier and Nagel (2016) show that inflation expectations track the inflation they have lived through, as if each individual has estimated an AR(1) on its own history, assigning higher weight to recent observations. I assume under the latter, the central bank can reduce the influence of lived inflation on the anchor

and displace it toward the inflation target. Assumption 3 formally states the two hypotheses.

ASSUMPTION 3 (Hypotheses About the Anchor). *Cohort g considers two values for the anchor,*

$$m_T = \pi^*, \qquad m_{E,g} = \nu\,\pi^* + (1-\nu)\,\bar{\pi}_g, \qquad \nu \in [0,1]. \tag{4}$$

Hypothesis T says the anchor is the target. Hypothesis E says the anchor is what the cohort lived through, moved toward the target by a fraction ν that measures the intensity of the central bank's target-communication. Finally, the lived inflation is given by $\bar{\pi}_g$.

To introduce central-bank communication into the formation of expectations, I focus on the content a central bank transmits when it talks about its inflation target. As documented by Binder (2017) and Lamla and Vinogradov (2019), policy announcements are generally unnoticed by the general public. However, an active target-communication campaign can succeed in shaping perceptions where passive announcement fails. For instance, Ehrmann et al. (2025) documented that the announcement of the ECB's 2021 monetary strategy review and the revised formulation of the inflation target went largely unnoticed by households. Despite household inattention, when the same content was delivered directly through randomized information treatments, the perceived credibility that price stability would be maintained increased significantly. Importantly, this paper does not provide an answer to the factors behind the apparent inattention or lack of interest by the public when it comes to target communication. Instead, it assumes that the central bank can reach households if it intentionally decides to communicate its target intensively.

Accordingly, I assume the central bank directly chooses the *intensity* of its target-communication through $\nu \in [0,1]$, where $\nu = 0$ is the baseline case in which the central bank is silent about the target and $\nu = 1$ is a target-communication strategy at its maximal intensity. Two properties of ν are worth mentioning. First, even when ν is a common communication strategy, its effects are cohort-specific, since it acts on each cohort's own experienced gap. Second, the intensity ν does not change the target or the lived inflation, but it changes what the cohort would believe under hypothesis *E*. Intuitively, a household that is consistently told what the inflation target is incorporates it into its alternative hypothesis, so its experiential anchor is pulled toward π^* by the fraction ν. Thus, under maximal target

communication the two hypotheses coincide at the target, and credibility becomes irrelevant for the forecast.

Households do not know which hypothesis is right and learn about it from the forecast errors of each one of them:

$$e_{h,t} = \pi_t - \rho_g \pi_{t-1} - (1 - \rho_g) m_h. \tag{5}$$

Individuals incorporate hypothesis errors into future inflation beliefs via Bayesian updating. Hence, individuals calculate the log likelihood of each observation under each hypothesis given those errors as

$$\log f(\pi_t \mid \pi_{t-1}, h) = -\tfrac{1}{2} \log\left(2\pi\sigma_g^2\right) - \frac{e_{h,t}^2}{2\sigma_g^2}. \tag{6}$$

The learning occurs via discounting old evidence at a geometric rate ρ_U. This is the "forgetting" factor of Koop and Korobilis (2012) and essentially measures how much past performance is weighted relative to recent performance when evaluating hypothesis h. Besides, I explicity make the assumption that Assumption 4 formalise that idea.

ASSUMPTION 4 (Discounted Learning). *Cohort g starts with prior probability* 1/2 *on each hypothesis and updates by Bayes' rule with the discounted likelihood of each hypothesis $h \in \{T, E\}$:*

$$\ell_{h,t} = \sum_{j \geq 0} \rho_U^j \log f(\pi_{t-j} \mid \pi_{t-j-1}, h), \qquad \rho_U \in (0, 1). \tag{7}$$

Now, consider hypothesis T (the anchor is the inflation target) and calculate its posterior using Bayes' rule, and assume that ex-ante, before any error is accumulated, the prior the agent assigns to each hypothesis is 0.5, so there is no ex-ante bias across hypotheses. Also, following Koop and Korobilis (2012), assume that only the errors observed up to time $t-1$ are considered when forecasting at time t. I define the outcome of hypothesis T posterior updating as the *subjective credibility* and is summarized by Proposition 1.

PROPOSITION 1 (Subjective Credibility). *Under Assumptions 1–4, the probability that*

cohort g attaches to hypothesis T is

$$\chi_{g,t} = \frac{e^{bU_{T,g,t-1}}}{e^{bU_{T,g,t-1}} + e^{bU_{E,g,t-1}}} = \left[1 + e^{-b\,(U_{T,g,t-1} - U_{E,g,t-1})}\right]^{-1}, \qquad b = \frac{1}{2\sigma_g^2\,(1-\rho_U)}. \tag{8}$$

where the performance $U_{h,t}$ of hypothesis $h \in \{T, E\}$ is

$$U_{h,t} = -(1-\rho_U)\,e_{h,t}^2 + \rho_U\,U_{h,t-1}. \tag{9}$$

PROOF. See Appendix A, subsection A.1. □

There are two immediate consequences of Proposition 1 that are closely related to the credibility proportion of Hommes and Lustenhouwer (2019). First, the posterior collapses to a logistic function of the difference in performance between hypotheses, which is why credibility takes the discrete-choice form as in their case. Second, the sensitivity parameter, b, contrary to their free "intensity of choice", here is the precision of inflation innovations corrected for forgetting. Hence, when inflation is noisy (high σ_g^2), a given forecast error is weak evidence against either hypothesis, and credibility moves slowly, but when inflation is predictable (low σ_g^2), the same forecast error moves credibility fast.

The next step is to calculate the predictive distribution of π_{t+1} weighted by credibility. I adopt the idea of averaging models that hold in different periods of time based on the "dynamic model averaging" of Koop and Korobilis (2012). Importantly, this approach incorporates credibility as part of the predictive distribution of each agent by itself and not as a proportion of the population as in Hommes and Lustenhouwer (2019). Therefore, the predictive distribution from averaging both hypotheses can be calculated, and directly its associated expectation, as Lemma 1 states.

LEMMA 1 (Predictive distribution and the posterior-mean anchor). *Under Assumptions 1–4, cohort g's predictive distribution for π_{t+1} is the mixture*

$$\begin{aligned} f(\pi_{t+1} \mid I_t) &= \chi_{g,t}\,\varphi\big(\pi_{t+1};\,\mu_{T,t},\,\sigma_g^2\big) + (1-\chi_{g,t})\,\varphi\big(\pi_{t+1};\,\mu_{E,t},\,\sigma_g^2\big), \\ \mu_{h,t} &\equiv \rho_g \pi_t + (1-\rho_g) m_h, \end{aligned} \tag{10}$$

where $\varphi(x;\mu,\sigma^2)$ *is the normal density, and its mean is*

$$\begin{aligned}\mathbb{E}_{g,t}[\pi_{t+1}] &= \rho_g\,\pi_t + (1-\rho_g)\,m_{g,t},\\ m_{g,t} &\equiv \chi_{g,t}\,\pi^* + (1-\chi_{g,t})\,m_{E,g} \;=\; \pi^* + (1-\chi_{g,t})(1-\nu)\,\Delta_g.\end{aligned} \tag{11}$$

where the distance between remembered inflation and the target,

$$\Delta_g \;\equiv\; \bar{\pi}_g - \pi^*,$$

is the cohort's experienced gap.

PROOF. See Appendix A, subsection A.2. □

As Equation (11) shows, the inflation expectation is the perceived law with the unknown anchor replaced by its posterior mean $m_{g,t}$, which is the target plus the part of the experienced gap that has not been removed by either credibility or communication, and which is ultimately where the central bank can exert its influence. Inflation expectations inherit the uncertainty about the anchor: when $\chi_{g,t} = 1$ (the *perfect-credibility scenario*), the anchor collapses to the target π^* regardless of the communication intensity selected by the central bank. At the opposite extreme, when $\chi_{g,t} = 0$ (the *null-credibility scenario*), the anchor collapses to $m_{E,g}$: expectations are anchored by the cohort's experiential inflation rate modulated by central-bank communication.

The ultimate step involves distorting the credibility-weighted predictive distribution by DE. Section 2 documents that household forecasts under-react to inflation news. Following Bordalo et al. (2018), I model this by assuming that the cohort does not report the mean of its predictive distribution but its *diagnostic* version, which reweights each state by how much more likely it has become since last period. Two restrictions determine where the distortion acts. First, in line with the DE literature, Assumption 5 says that it acts on the forecast. I additionally assume that it takes credibility as a state that is already known, otherwise the operator would also distort average posterior. Thus, the same $\theta < 0$ that produces under-reaction in the forecast would make credibility under-react too, which would complicate interpretation since the credibility process would not be independently identified.

ASSUMPTION 5 (Diagnostic Distribution). *Cohort g does not report the mean of its*

predictive distribution. Instead, it reports its diagnostic *version, the distribution*

$$f^{\theta}(\pi_{t+1} \mid I_t) \;\propto\; f(\pi_{t+1} \mid I_t)\left[\frac{f(\pi_{t+1} \mid I_t)}{f(\pi_{t+1} \mid I_{t-1})}\right]^{\theta}. \tag{12}$$

which reweights each value of π_{t+1} by how much more likely it has become since last period.

The parameter θ is common across cohorts. With $\theta > 0$ the cohort overweights states that became more likely (over-reaction), while $\theta < 0$ means it underweights them (under-reaction). Therefore, the distortion is applied to the predictive distribution of inflation under uncertainty about the anchor and takes credibility as a state of the perceived law that is already known when the forecast is made. That is, inside the agent's mind, she first sets her subjective credibility and then reports her diagnostic distribution. Additionally, Assumption 6 says that DE does not act on the learning process, which implies that hypotheses are evaluated by their own raw errors, and not by the errors of the distorted forecast.

ASSUMPTION 6 (Learning Independent of DE). *The two hypotheses, $\{T, E\}$, are compared by the errors of their own forecasts under Equation* (3)*, not by the error of the diagnostic forecast, so the learning probabilities are not distorted.*

Following L'Huillier et al. (2024), direct application of the resulting diagnostic distribution of Equation (12) leads to:

$$\mathbb{E}_t^{\theta}[\pi_{t+1}] \;=\; \mathbb{E}_t[\pi_{t+1}] + \theta\big(\mathbb{E}_t[\pi_{t+1}] - \mathbb{E}_{t-1}[\pi_{t+1}]\big). \tag{13}$$

Proposition 2 linearize this equation to ultimately deliver an inflation expectation that endogenously captures credibility and under-reaction to news, and the expression next sections will be adopted throughout.

PROPOSITION 2 (Diagnostic Expectation with Endogenous Credibility). *FOr cohort g, the Inflation Expectations linearized around the steady state, $= \pi^*$, is*

$$\mathbb{E}_{g,t}^{\theta}[\hat{\pi}_{t+1}] \;=\; \rho_g\,\hat{\pi}_t + (1-\rho_g)\,\kappa_g\,\hat{\chi}_{g,t} + \theta\,\rho_g\big(\hat{\pi}_t - \rho_g\,\hat{\pi}_{t-1} - (1-\rho_g)\,\kappa_g\,\hat{\chi}_{g,t-1}\big). \tag{14}$$

where κ_g is the credibility sensitivity of the anchor,

$$\kappa_g \;=\; -(1-\nu)\,\Delta_g. \tag{15}$$

PROOF. See Appendix A, subsection A.4. □

The persistence ρ_g captures the idea that a cohort that lived through persistent inflation expects any inflationary shock to last longer, and revises its expectations more. On the other hand, the credibility sensitivity κ_g shows that cohorts whose experienced inflation is close to the target respond only slightly to credibility changes, whereas cohorts whose inflation experience is far from the target exhibit larger expectation adjustments when credibility changes.

4. Credibility-Distorted Beliefs and the New-Keynesian Model

This section embeds the credibility-distorted beliefs of Section 3 into a Representative Agent New-Keynesian (RANK) model. The belief distortion essentially operates through the cohort's perceived real interest rate and the aggregate Euler equation, which ultimately causes cohorts to deviate from the FIRE intertemporal consumption path. Next, I extend the analysis to a richer model of the economy by considering a lifecycle Heterogeneous-Agent New-Keynesian (HANK) model in which cohorts also differ in their marginal propensities to consume (MPC) and in their balance-sheet exposure (unhedged interest-rate exposure, URE). This step makes clear how distorted beliefs now interact with the redistribution channel of monetary policy. While the *perceived real rate* affects consumption by distorting the intertemporal consumption path as in the RANK model, the *realised real rate,* the interest rate that households actually earn on their bonds, revalues each cohort's balance sheet through the URE channel. Hence, this distinction shows that the incidence in consumption shifts from the cohorts that forecast differently, as in the RANK model, to the cohorts that spend differently, as in the HANK model.

4.1. The Distorted RANK Model

This simple model consists of a discrete-time New Keynesian economy populated by a continuum of households divided into G cohorts. Every group is indexed by $g \in \{1, \dots, G\}$ and has population weight $\omega_g \geq 0$, where $\sum_g \omega_g = 1$. Each cohort follows the same preferences over consumption $u(C) = C^{1-\sigma^{-1}}/(1-\sigma^{-1})$, derive disutility from labor $\zeta(N) = N^{1+\psi^{-1}}/(1+\psi^{-1})$, share the common income $y_{g,t} = y_t$, so there is no income heterogeneity. Households have access to a one-period nominal bond a_t paying the nominal interest rate i_t. Aggregate variables are

expressed in uppercase letters while lowercase letters are reserved to deviations from the flexible-price steady state. Finally, the goods market clearing condition imposes $y_t = \int c_{i,t} \mathrm{d}i = \sum_g \omega_g c_{g,t}$, with $c_{g,t} \equiv \int_{i \in g} \frac{1}{\omega_g} c_{i,t} \mathrm{d}i$.

To cleanly isolate how the beliefs distorted channel shapes inflation expectations, I assume the beliefs about future output for cohort g are FIRE so, $\mathbb{E}_{g,t}[y_{t+h}] = y_{t+h}$ holds and anchored-recall only operates through inflation expectations and hence through the perceived real rate. While it may be possible that the distorted expectations developed affects how agents forecast future output, the empirical strategy used in Section 2 with the BoE Inflation Attitudes Survey does not have it counterpart for output expectations, so imposing a parallel channel on y would imply accepting a non-analyzed mechanism, which is out of the scope of this paper.

Each cohort g solves the maximization problem:

$$\max_{\{C_{g,t}, a_{g,t+1}, N_{g,t}\}_{t \geq 0}} \mathbb{E}^{\theta}_{g,0} \sum_{t=0}^{\infty} \beta_g^t \left\{ u(C_{g,t}) - \zeta(N_{g,t}) \right\} \tag{16}$$

subject to

$$C_{g,t} + a_{g,t+1} = Y_t + \left(1 + r_t\right) a_{g,t}, \tag{17}$$

and the no-Ponzi condition $\lim_{T \to \infty} \beta_g^T a_{g,T} \geq 0$ holds, which binds with equality at the optimum (transversality).

Importantly, Proposition 2 implies that for every horizon $h \geq 0$, cohort g's perceived ex-ante real rate is

$$r^{\text{per}}_{g,t+h} = i_{t+h} - \rho_g \pi_{t+h} - (1 - \rho_g) m_{g,t+h} - \theta \rho_g \left[u_{t+h} - (1 - \rho_g) m_{g,t+h-1} \right]. \tag{18}$$

In this simple version of the economy, absent of wealth accumulation, the realised real rate, r_t, only affects the return of current assets and not intertemporal consumption decisions, which depend on the perceived rate $r^{per}_{g,t}$ alone. Moreover, since cohorts hold no assets in steady state, $a^{ss}_g = 0$, there is no stock of wealth to be revalued when r_t moves, so the realised rate has no effect on consumption. This is the key difference with the HANK model, where there are heterogeneous asset positions, so the same movement in r_t revalues their cash-on-hand, the URE channel.

Intertemporal consumption decisions are pinned down by the perceived real rate of cohort g. This rate depicts the cohort-specific channels through which

the distortion enters into the consumption-saving decision. The first mechanism states that cohorts that lived through persistent inflation (high ρ_g) perceive a lower real rate, whereas low-persistence cohorts perceive a real rate closer to the nominal rate i_{t+h}. The second channel propagates central-bank credibility $\chi_{g,t+h}$ through $\kappa_g = \pi^* - m_{E,g}$. It asserts that cohorts whose experienced inflation has persistently diverged from the target react more strongly to credibility changes. Finally, the diagnostic parameter θ measures how the cohort reacts to news. When $\theta < 0$, the cohort under-reacts to news, so the perceived real rate falls by less than what the inflation news would demand, therefore households bring consumption forward by a lower magnitude of what the FIRE case would justify.

The standard solution of the household maximization problem leads to the cohort-g distorted Euler equation:

$$C_{g,t}^{-1/\sigma} \;=\; \beta_g (1 + r_{g,t}^{\text{per}})\, \mathbb{E}_{g,t}^{\theta}[C_{g,t+1}^{-1/\sigma}] \tag{19}$$

It is noteworthy mentioning the degree of cohort heterogeneity introduced by β_g, permitting the differences across cohorts in the perceived real interest rate will prevail in steady-state, $r_g^{per\,ss} = \beta_g^{-1} - 1$. The linearized version around the steady state for the perceived rate obeys

$$\hat{r}_{g,t+h}^{\text{per}} \;=\; \hat{\imath}_{t+h} - \rho_g\, \hat{\pi}_{t+h} - (1-\rho_g)\, \kappa_g\, \hat{\chi}_{g,t+h} - \theta\, \rho_g \big(\hat{\pi}_{t+h} - \rho_g\, \hat{\pi}_{t+h-1} - (1-\rho_g)\, \kappa_g\, \hat{\chi}_{g,t+h-1}\big). \tag{20}$$

while, Lemma 2 shows the equivalent expression for consumption:

LEMMA 2 (The cohort-g log-linearized consumption). *The first-order log-linearised solution of* (16)–(17) *around* $a_g^{ss} = 0$, $\pi^{ss} = \pi^*$, χ_g^{ss}, *and* $r_g^{per\,ss} = \beta_g^{-1} - 1$ *is*

$$c_{g,t} \;=\; (1 - \beta_g) \sum_{h=0}^{\infty} \beta_g^h\, y_{t+h} \;-\; \sigma \sum_{h=0}^{\infty} \beta_g^{h+1}\, \hat{r}_{g,t+h}^{per}, \tag{21}$$

where $\hat{r}_{g,t+h}^{per}$ *is given by* (20).

PROOF. See Appendix A, subsection A.5. □

As a final step, assume that all cohorts smooth consumption at roughly the same speed, which implies $\beta_g^h \approx \bar{\beta}^h$ for all g, h, and define $\bar{\beta} \equiv \sum_g \omega_g \beta_g$, the population-average perceived rate $\bar{r}_t^{\text{per}} \equiv \sum_g \omega_g \hat{r}_{g,t}^{\text{per}}$, and recall the goods-market

clearing condition $y_t = \sum_g \omega_g c_{g,t}$. Then, Equation 21 may be simplified to:

$$y_t \;=\; (1-\bar{\beta}) \sum_{h=0}^{\infty} \bar{\beta}^h \, y_{t+h} \;-\; \sigma \sum_{h=0}^{\infty} \bar{\beta}^{h+1} \, \bar{r}^{\text{per}}_{t+h}. \tag{22}$$

Cancelling common terms ultimately leads to:

$$y_t \;=\; y_{t+1} \;-\; \sigma \, \bar{r}^{\text{per}}_t .$$

Thus, the aggregation of the cohort-g Euler equations reduces to the representative-agent Euler equation.

To close the model, firms set prices à la Calvo and have perfect-foresight which yields the standard New Keynesian Phillips curve, augmented with an exogenous cost-push shock ε^{CP}_t that follows an AR(1) process with persistence ρ. The central bank conducts monetary policy through a Taylor rule that responds to inflation and is subject to an exogenous shock ε_t. Together with the aggregate Euler equation derived previously, the equilibrium of this simple economy is characterized by the following three-equation block:

$$\text{Policy Rule:} \quad i_t = \phi \pi_t + \varepsilon_t, \tag{23}$$

$$\text{NKPC:} \quad \pi_t = \varpi \, y_t + \beta \, \mathbb{E}_t[\pi_{t+1}] + \varepsilon^{CP}_t, \tag{24}$$

$$\text{Aggregate Euler:} \quad y_t = \mathbb{E}_t[y_{t+1}] - \sigma \, \bar{r}^{\text{per}}_t, \qquad \bar{r}^{\text{per}}_t \equiv \sum_g \omega_g \, \hat{r}^{\text{per}}_{g,t} . \tag{25}$$

The three-equation block (23)–(25), together with the distorted expectations of Proposition 2, is the system that the RANK simulations solve. Importantly for the welfare analysis developed later in the paper, cohorts react more or less strongly than the average, depending on their own distortion parameters ρ_g and κ_g, through their perceived real interest rate. To bring the model closer to the HANK literature, the next step is to introduce uninsurable income risk and heterogeneous asset positions, so that cohorts would differ in their marginal propensities to consume (MPCs) and in the exposure (URE) of their balance sheets to r_t.

4.2. The Distorted HANK Model

This model builds on the lifecycle HANK of Auclert et al. (2024), augmented with the distorted expectations mechanism introduced in Section 3 and the cohort belief

heterogeneity structure of Guerreiro (2023). The economy consists of a continuum of infinitely-lived households $i \in [0,1]$, where each household belongs to an age-cohort $g \in \{1,\dots,G\}$. Every cohort has a population weight $\omega_g \geq 0$, $\sum_g \omega_g = 1$ and a discount factor $\beta_g \in (0,1)$. As standard in the HANK literature, I assume that households face uninsurable idiosyncratic labour-productivity risk $e_{i,t}$, which follows a Markov chain with transition matrix $P(e,e')$. This matrix is common across cohorts and has stationary distribution $\pi^e(e)$ such that $\sum_e \pi^e(e)\, e = 1$.

Every household in group g receives a post-tax labour income from supplying $N_{g,t}$ hours of labor (determined by unions) given by:

$$y_{g,t}(e) \;=\; (1-\tau_t)\, e\, w_t\, N_{g,t}, \tag{26}$$

where w_t is the real wage and τ_t the labour-income tax. Finally, markets are incomplete and households trade a one-period real bond a such that $a \geq 0$.

To keep consistency with Proposition 2, I introduce the diagnostic-expectations mechanism *only* through distorted beliefs about the aggregate inflation path, which in turn shape the perceived real interest rate that drives consumption decisions through the *Fisher channel*. By contrast, I assume that households observe their idiosyncratic state $e_{i,t}$ and are fully rational with respect to it. Moreover, they know the transition matrix $P(e,e')$ and form correct expectations about their own income. Accordingly, the cohort-g perceived ex-ante real rate is,

$$r^{\text{per}}_{g,t} \;\equiv\; i_t - \mathbb{E}^{\theta}_{g,t}[\pi_{t+1}]. \tag{27}$$

Importantly, distorted beliefs operate only at the cohort level. The real return that wealth actually earns is the *realised* ex-post real rate, $r_{t+1} = i_t - \pi_{t+1}$. Thus, a bond's realised return does not depend on what its holder believes, so every household earns r_{t+1} on its bonds *regardless of its expectations*. The cohort-perceived rate $r^{\text{per}}_{g,t}$ in (27) is the belief about that same return held at the decision date, namely the Fisher equation evaluated under the diagnostic expectation $\mathbb{E}^{\theta}_{g,t}$, and it is the rate that *shapes inter-temporal consumption decisions*. The two rates therefore coexist as a realised return and a forecast of it. This distinction does not emerge in the RANK model, where households hold no wealth in steady state, so the realised rate has nothing to revalue and only $r^{\text{per}}_{g,t}$ enters the Euler equation, as discussed before.

For every cohort-g household entering period t with bond holdings a and

productivity e, the consumption and savings decisions are obtained from solving the Bellman equation

$$\begin{aligned} V_{g,t}(a,e) \;=\; \max_{c,\,a'\geq 0} \; u(c) \;+\; \beta_g\, \mathbb{E}^{\theta}_{g,t}\!\left[\, V_{g,t+1}(a',e') \,\middle|\, e \,\right] \\ \text{s.t.} \quad c + a' \;=\; y_{g,t}(e) \;+\; \left(1 + r_t\right) a, \end{aligned} \tag{28}$$

where $u(c) = c^{1-\sigma^{-1}}/(1-\sigma^{-1})$. The solution to Problem (28) defines the cohort consumption and asset policy functions $c_{g,t}(a,e)$ and $a_{g,t}(a,e)$.

Remarkably, the budget constraint in (28) uses r_t on existing wealth, while the optimal saving decision discounts future marginal utility at the cohort-perceived rate. This implies the distorted Euler equation:

$$u'(c_{g,t}) \;=\; \beta_g \left(1 + r^{\text{per}}_{g,t}\right) \mathbb{E}^{\theta}_{g,t}\!\left[\, u'(c_{g,t+1}) \,\right], \tag{29}$$

so the discount rate between t and $t+1$ uses the nominal rate and the diagnostic expectation of the decision date t, exactly as in the model of Section 4.1.

As discussed in Guerreiro (2023), Problem (28) imposes the law of iterated expectations at the individual level, $\mathbb{E}^{\theta}_{g,t}[\mathbb{E}^{\theta}_{g,t+1}[\cdot]] = \mathbb{E}^{\theta}_{g,t}[\cdot]$, so households behave as if they do not expect to revise their inflation beliefs. Since diagnostic expectations do not distort the idiosyncratic state, the cross-sectional distribution of beliefs is orthogonal to (a,e). This condition allows the behavioural Jacobians of Appendix A.6 to be computed from their FIRE counterparts.

Finally, denote by $D_{g,t}(a,e)$ the joint distribution of assets and productivity. Given the policy $a_{g,t}(\cdot)$ and the exogenous chain P, it evolves according to[5]

$$D_{g,t+1}(a',e') \;=\; \sum_{e} \int \mathbf{1}\!\left\{ a_{g,t}(a,e) = a' \right\} P(e,e')\; D_{g,t}(\mathrm{d}a, e). \tag{30}$$

Hence, cohort aggregates may be calculated as

$$C_{g,t} \;=\; \sum_{e} \int c_{g,t}(a,e)\, D_{g,t}(\mathrm{d}a,e), \qquad A_{g,t} \;=\; \sum_{e} \int a_{g,t}(a,e)\, D_{g,t}(\mathrm{d}a,e), \tag{31}$$

while aggregate consumption and bond demand are given by $C_t = \sum_g \omega_g C_{g,t}$ and $A_t = \sum_g \omega_g A_{g,t}$.

[5] I implement the lottery method described in Young (2010) on a grid, so (30) becomes $D_{g,t+1} = \Lambda'_{g,t}\, D_{g,t}$ for a sparse transition matrix $\Lambda_{g,t}$.

The rest of the model is standard. The supply side of the economy, common across cohorts, follows Guerreiro (2023) and is the canonical sticky-wage New Keynesian block of Auclert et al. (2024). A competitive final-good firm operates the linear technology $Y_t = \Theta_t N_t$, so the real wage equals marginal productivity, $w_t = \Theta_t$. Price inflation is therefore wage inflation net of productivity growth. As in Guerreiro (2023), a continuum of unions aggregates cohort labour into N_t, sets wages subject to a quadratic adjustment cost, and delivers the linearised wage Phillips curve

$$\pi_t^w \;=\; \kappa_w\Big[\sigma^{-1}\hat{C}_t + \psi^{-1}\hat{N}_t - \big(\hat{w}_t - \hat{\tau}_t\big)\Big] \;+\; \bar{\beta}\,\mathbb{E}_t[\pi_{t+1}^w] \;+\; \varepsilon_t^{CP}, \tag{32}$$

with κ_w the slope, ψ the Frisch elasticity, ε_t^{CP} an exogenous cost-push shock, and hats denoting log-deviations. Cohort hours are allocated by the uniform union rule $N_{g,t} = \Gamma_g(N_t)$, with $\sum_g \omega_g \Gamma_g(N_t) = N_t$. Unions set wages and ration hours uniformly across households, so households do not choose labour directly and N does not enter their intertemporal problem. The government issues real debt $B_t = B$, spends $G_t = G$, and adjusts τ_t to satisfy $G + (1 + r_t)B = \tau_t Y_t + B$, and monetary policy follows a Taylor rule with an exogenous monetary shock ε_t^{MP},

$$i_t = r_{ss} + \phi_\pi \pi_t + \varepsilon_t^{MP}, \qquad r_{t+1} = i_t - \pi_{t+1}. \tag{33}$$

Finally, the equilibrium for this economy is defined in Appendix B.

4.2.1. Sequence-Space Representation

This paper uses the sequence space approach introduced by Auclert et al. (2021) due to its effectiveness for solving heterogeneous-agent models when nonlinearities do not arise. In order to obtain the first-order transition dynamics, the economy is written as a set of blocks, each being fed by inputs to produce time paths of aggregate outputs. Define the first-order behaviour of a generic block with output O and input X as its Jacobian matrix

$$\mathcal{J}^{O,X} \;=\; \Big[\,\partial O_t / \partial X_s\,\Big]_{t,s\geq 0}, \tag{34}$$

whose (t, s) entry captures how the date-t output responds to a unit innovation in the date-s input, evaluated at the steady state. Intuitively, column s will be interpreted as the impulse response of O to news, received today, about the date-s

value of X. Following Auclert et al. (2021), the household blocks' Jacobians are computed with the fake-news algorithm.

The economy described previously contains G cohort household blocks plus the production, fiscal, bond-pricing, Phillips-curve, monetary-policy, diagnostic-expectations, credibility, and market-clearing blocks. Any block other than the g household problems is a simple analytical block that resembles Galí (2015). Each cohort g block is a standard incomplete-markets recursion, and these are the only non-trivial blocks.[6]

Through the functional $\mathcal{C}_g(\cdot)$, cohort g takes as inputs the realised real rate r_t, the perceived real rate $r_{g,t}^{\text{per}}$ of (27), and its after-tax labour income $(1-\tau_t)\, w_t N_{g,t}$, and returns as outputs cohort consumption $C_{g,t}$ and asset holdings $A_{g,t}$ with

$$\mathcal{C}_g\big(\{r_t, r_{g,t}^{\text{per}}, Y_{g,t}, \tau_t\}_{t\geq 0}\big) \;=\; \sum_e \int c_{g,t}(a,e)\, D_{g,t}(\mathrm{d}a, e), \tag{35}$$

and analogously for asset demand $\mathcal{A}_g$. The same block returns each cohort's consumption, with $\sum_g \omega_g \mathcal{C}_g = C$, used to calculate the cohort-level wedges for the HANK model in the next section.

The first-order response of cohort consumption $\mathbf{C}_g$ is a linear combination of its inputs' Jacobian contributions.[7]

$$\mathrm{d}\mathbf{C}_g \;=\; \mathcal{J}_g^{C,r} \cdot \mathrm{d}\mathbf{r} \;+\; \mathcal{J}_g^{C,r^{\text{per}}} \cdot \mathrm{d}\mathbf{r}_g^{\text{per}} \;+\; \mathcal{J}_g^{C,Y} \cdot \mathrm{d}\mathbf{Y}_g \;+\; \mathcal{J}_g^{C,\tau} \cdot \mathrm{d}\boldsymbol{\tau}. \tag{36}$$

The Jacobian $\mathcal{J}_g^{C,r}$ is the income (cash-on-hand) consumption response to the realised real rate. This object matters because it captures the pure redistribution channel of the realised interest rate, whose impact response across cohorts reproduces the unhedged interest-rate exposure (URE) of Auclert (2019). Intuitively, the URE measures each household's net maturing position and its refinancing needs

[6] I adopt the same solution framework as in Auclert et al. (2021). Given the paths of inputs for every heterogeneous g block, I solve the problem (28) backwards for the marginal value of wealth and the consumption and savings policies $c_{g,t}(a,e)$, $a_{g,t}(a,e)$ using the endogenous-gridpoint method, over a state of idiosyncratic productivity e (evolving on the Markov chain $P(e,e')$) and bond holdings a on a grid. Then, I iterate the cohort distribution $D_{g,t}(\mathrm{d}a, e)$ forward and aggregate the policies. Cohorts follow the same recursion and the only differences are the discount factor β_g, the labour income $y_{g,t}$, and the associated perceived rate $r_{g,t}^{\text{per}}$. Finally, time paths are truncated at a horizon T, and the household Jacobians are computed once at the steady state.

[7] In terms of notation, bold letters denote the stacked column vector of deviations at the different horizons, $\mathbf{X} = (X_0, X_1, \dots)'$.

every period:

$$\text{URE} = \text{income} + \text{maturing assets} - \text{consumption} - \text{maturing debt}.$$

A household with $\text{URE}_g > 0$ is a short-term net saver, namely, a rise in the real rate raises its cash-on-hand, since it reinvests maturing positions at the higher rate, and it increases consumption. Conversely, a household with $\text{URE}_g < 0$ is a short-term net borrower since the same rise lowers its cash-on-hand, as it must refinance at the higher rate, and reduces consumption. Auclert (2019) argues that URE is a sufficient statistic to determine the winners and losers of any interest rate change since $\mathrm{d}C^r = \sum_g \omega_g \,\text{MPC}_g \,\text{URE}_g \,\mathrm{d}r$. Hence, only the joint distribution of marginal propensities and exposures matters to determine this consumption response. Since young cohorts, which hold the smallest asset positions, have the highest MPC, this covariance is negative and the redistribution channel amplifies monetary transmission in their model.

Distorted beliefs introduce an additional mechanism through which the central bank's decisions may influence consumption. Expanding $\mathrm{d}\mathbf{r}_g^{\text{per}}$ in Equation 36 gives:

$$\begin{aligned}\mathcal{J}_g^{C,r^{\text{per}}}\mathrm{d}\mathbf{r}_g^{\text{per}} = \underbrace{\mathcal{J}_g^{C,r^{\text{per}}}\mathrm{d}\mathbf{i}}_{\text{Fisher}} - \underbrace{\rho_g\,\mathcal{J}_g^{C,r^{\text{per}}}\big[(1+\theta)\mathbb{I} - \theta\rho_g L\big]\mathrm{d}\boldsymbol{\pi}}_{\text{Inflation}} \\ \underbrace{-\,(1-\rho_g)\,\kappa_g\,\mathcal{J}_g^{C,r^{\text{per}}}\big[\mathbb{I} - \theta\rho_g L\big]\mathrm{d}\boldsymbol{\chi}_g}_{\text{Credibility}}.\end{aligned} \tag{37}$$

Equation 37 shows that, with L the lag operator, the perceived-rate Jacobian decomposes into a pure Fisher channel, an inflation component $\rho_g[(1+\theta)\mathbb{I} - \theta\rho_g L]\mathcal{J}_g^{C,r^{\text{per}}}$, and a credibility component $(1-\rho_g)\kappa_g[\mathbb{I} - \theta\rho_g L]\mathcal{J}_g^{C,r^{\text{per}}}$. The last one is the only channel that central-bank communication can influence, since its communication ν enters only through $m_{E,g}$.

To close the model, note that the cohort household blocks, the supply block (the Phillips curve), and the policy block jointly determine the aggregate paths $\mathbf{U} = (\mathbf{Y}, \boldsymbol{\pi}, \boldsymbol{\chi})'$ through the equilibrium conditions (32)–(A30). The model is solved in sequence space, taking output, inflation and credibility as unknowns, and imposing the asset-market clearing, the Phillips curve and the credibility law of motion conditions,

$$H\big(\mathbf{U},\mathbf{Z}\big)=\mathbf{0},\qquad \mathbf{Z}=(\varepsilon^{CP})',\tag{38}$$

where $\mathbf{Z}$ collects the cost-push shock, ε^{CP}. Linearising around the steady state, the response of the aggregates to any shock is

$$\mathrm{d}\mathbf{U}=-H_{\mathbf{U}}^{-1}H_{\mathbf{Z}}\,\mathrm{d}\mathbf{Z},\tag{39}$$

where $H_{\mathbf{U}}$ and $H_{\mathbf{Z}}$ are the corresponding Jacobians. The belief distortion enters this system only through the cohort household blocks, via their perceived real rate $r_{g,t}^{\text{per}}$ of (27). Therefore, a single Jacobian, $\mathcal{J}_g^{C,r^{\text{per}}}$, is enough to carry the inflation and the credibility components of (37) to consumption. Hence, no additional belief Jacobian or forecast-revision term is required, since all learning dynamics may be obtained from the credibility path χ_g. This is the representation that Proposition 4 uses in the next section to compute the cohort consumption wedges.

Finally, Table 1 reports the calibration. Panel A collects the structural parameters of the three-equation block (23)–(25), set to the standard values of Auclert et al. (2024), together with the diagnostic parameter θ, imported from Carroll (2003). Since the DE operator imposes $\mathbb{E}^{\theta}_{g,t}[\pi_{t+1}]=\mathbb{E}_{g,t-1}[\pi_{t+1}]+(1+\theta)\big(\mathbb{E}_{g,t}[\pi_{t+1}]-\mathbb{E}_{g,t-1}[\pi_{t+1}]\big)$, the diagnostic expectation incorporates a fraction $1+\theta$ of the forecast revision on impact, which matches the share $\lambda=0.27$ of households that update each quarter in Carroll (2003), so that $\theta=\lambda-1=-0.73$. The learning parameters ρ_U and b complete the panel. Both are imported from Hommes and Lustenhouwer (2019): the memory of past forecast errors is $\rho_U=0.5$ and the learning intensity is $b=0.25$. Panel B builds the cohort moments from UK data: the demographic weights ω_g are UK ONS mid-2019 shares, while the lifetime inflation persistence ρ_g and experienced mean $\bar{\pi}_g$ are constructed from the ONS quarterly-annualised CPI series (Appendix C.6.1), measured up to 2019Q4. It makes evident the heterogeneity across cohorts in experienced inflation. This experience determines how each cohort's anchor compares to the target, and it is ultimately the source of cohort heterogeneity when the central bank implements its communication strategy.

TABLE 1. **Calibration of the simple model.**

Panel A. Structural and belief parameters

Symbol	Value	Description	Source
β	0.99	Discount factor, quarterly	Auclert et al. (2024)
σ	1.00	Intertemporal elasticity of substitution	Auclert et al. (2024)
ϖ	0.03	New Keynesian Phillips curve slope	Auclert et al. (2024)
ϕ	1.50	Taylor-rule coefficient on inflation	Auclert et al. (2024)
ρ	0.50	AR(1) persistence of the shocks ε_t, ε_t^{CP}	Galí (2015)
π^*	2.00%	Inflation target	
θ	−0.73	Diagnostic parameter	Carroll (2003)
ρ_U	0.50	Memory of the discounted likelihood	Hommes and Lustenhouwer (2019)
b	0.25	Learning intensity	Hommes and Lustenhouwer (2019)
ν	0.00	Intensity of target-communication	
T	40	Horizon for the IRFs (quarters)	

Panel B. Cohort-level calibration

	15–24	25–34	35–44	45–54	55–64	65+	Pop. mean
Population weight, ω_g	0.148	0.163	0.153	0.168	0.146	0.222	1.000
Inflation persistence, ρ_g	0.113	0.106	0.035	0.179	0.559	0.554	0.274
Experienced inflation, $\bar{\pi}_g$ (%)	2.02	2.35	2.00	2.66	4.21	5.04	3.17

Notes: ρ_g and $\bar{\pi}_g$ are the AR(1) slope and the mean of the cohort's lifetime quarterly-annualised UK CPI inflation up to 2019Q4. The last column of Panel B reports population-weighted means.

5. Welfare and Central Bank Communication

This section makes the case that central bank communication has welfare and distributional effects when beliefs are distorted. In the RANK model with FIRE, every cohort forecasts future inflation correctly and picks the consumption path it would choose with perfect foresight, so there is no welfare loss from beliefs. However, under distorted beliefs this is no longer true. Each cohort's experienced inflation and credibility shape its forecast of inflation and its perceived real interest rate. This heterogeneity in Euler equations is what ultimately changes each cohort's consumption path, since each cohort departs from the path it would choose if it forecast inflation correctly. According to Gabaix (2020) this deviation is a valid welfare measure since it is evaluated with objective, rather than subjective

expectations. In this sense, the cost associated associated to a belief distortion is the utility lost along the consumption path it induces, relative to the path the household would choose with correct beliefs, justifying why this paper adopts it as measure of welfare throughout.

5.1. Consumption Wedges

Define the *rational path*, $dc^R_{g,t}$, as the consumption path cohort g would choose for the same realised paths of prices and income, under perfect foresight of inflation. Hence, the rational path measures the incidence of the shock with correct beliefs, and the gap between the actual path and the rational path measures the cost of distorted beliefs. However, the reference path behaves differently across models. In the RANK model the only thing that changes across cohorts is inflation expectations, so the rational path is common to all cohorts. On the other hand, in the HANK model the rational path is cohort specific, because even if a cohort has correct beliefs its consumption path responds to its idiosyncratic factors, such as the effect of the realised real rate through its asset position, and of its labour income and taxes through its MPC.

TABLE 2. **Variance Decomposition of the Cohort Rational Consumption Path**

	RANK	HANK					
Channel (share of $\sum_t (dc^R_{g,t})^2$, %)	(common)	15–24	25–34	35–44	45–54	55–64	65+
Nominal rate	150.2	34.6	36.6	39.3	43.0	45.6	46.6
Expected inflation (correct beliefs)	−50.2	−9.5	−10.1	−10.8	−11.9	−12.7	−13.0
Real rate (substitution)	100.0	25.0	26.5	28.5	31.0	32.9	33.7
URE (real return on assets)	0.0	−3.0	−3.2	−3.5	−3.7	−3.9	−4.0
Earnings heterogeneity	0.0	78.0	76.7	75.0	72.7	71.0	70.4
Total	100.0	100.0	100.0	100.0	100.0	100.0	100.0

Table 2 shows the variance decomposition of the cohort consumption response to a +1pp cost-push shock. Importantly, in the RANK model beliefs explain a considerable amount of the consumption response, and the rest is the real-rate channel. However, in the HANK model the effect of beliefs almost disappears and the rest is divided between the real-rate channel and the earnings-heterogeneity channel. The latter dominates because the cohorts with the highest MPC are also

the ones that depend most on labour income, which is the earnings-heterogeneity channel documented by Auclert (2019). Therefore, one caveat is that even when beliefs are a small part of the consumption response, they are the only part the central bank can reach with communication.

Belief heterogeneity makes each cohort depart from the consumption path it would choose with correct beliefs, and that deviation is costly. Define the cohort-g *consumption wedge* as the difference between the consumption path cohort g chooses under its *own* distorted belief and its *rational path*:

$$\mathcal{W}^R_{g,t} \;\equiv\; dc_{g,t} - dc^R_{g,t}. \tag{40}$$

Similarly, define the *consensus error* as the population average of these consumption wedges:

$$\Delta^R_t \;\equiv\; \sum_g \omega_g \, \mathcal{W}^R_{g,t} \;=\; \overline{dc}_t - \overline{dc^R}_t. \tag{41}$$

Importantly, consumption wedges isolate the effect of distorted beliefs on consumption and they are zero when the cohort forecasts correctly. Once the equilibrium paths of inflation and credibility are known, the wedges follow immediately. Next, define the *cohort-g* and *aggregate* welfare losses, respectively, as:

$$L^R_{g,t} \;\equiv\; \tfrac{1}{2}\big(\mathcal{W}^R_{g,t}\big)^2, \qquad \mathcal{L}^R_t \;\equiv\; \sum_g \omega_g \, L^R_{g,t}. \tag{42}$$

Proposition 3 characterises the consumption wedges and welfare losses in the RANK model, where only beliefs differ across cohorts, and Proposition 4 does so in the HANK model, where balance sheets and income differ as well.

PROPOSITION 3 (Consumption Wedges and Aggregate Welfare in RANK). *Consider any sequence of structural shocks that produces equilibrium paths* $\{\hat{\imath}_t, \hat{\pi}_t, \hat{\chi}_{g,t}\}_{t\geq 0}$ *of the RANK model in equations* (23)–(25). *Define* $dc_{g,t}$ *as the cohort-g consumption path that satisfies its own distorted Euler equation, and* dc^R_t *as the* ***rational path****: the consumption path a cohort would choose on the same realised paths with perfect foresight of inflation, so that its perceived real rate is the realised one,* $\hat{r}^R_t \equiv \hat{\imath}_t - \hat{\pi}_{t+1}$*, and* $dc^R_t = -\sigma \sum_{s\geq 0} \hat{r}^R_{t+s}$ *is common to all cohorts. Denote* $\overline{dc}_t \equiv \sum_g \omega_g \, dc_{g,t}$ *as the equilibrium aggregate path. The linearized inflation expectations follows* $\mathrm{d}\mathbb{E}^{\theta}_{g,t}[\hat{\pi}_{t+1}] = \Lambda^{\pi}_g \hat{\pi}_t + \Lambda^{\chi}_g \hat{\chi}_{g,t}$ *with*

$$\Lambda^{\pi}_g \hat{\pi}_t \;\equiv\; \rho_g\big[(1+\theta)\hat{\pi}_t - \theta\rho_g \hat{\pi}_{t-1}\big], \qquad \Lambda^{\chi}_g \hat{\chi}_{g,t} \;\equiv\; (1-\rho_g)\,\kappa_g\big[\hat{\chi}_{g,t} - \theta\rho_g \hat{\chi}_{g,t-1}\big], \tag{43}$$

Let $A_g \equiv -U_g''(c^R_{g,t}) > 0$ denote cohort g's coefficient of absolute risk aversion at the rational path. Then, the ***cohort-g consumption wedge****, $\mathcal{W}^R_{g,t} \equiv dc_{g,t} - dc^R_t$, is the cohort's cumulated one-step-ahead forecast error and splits exactly into a* ***diagnostic error*** *and a* ***credibility error****,*

$$\mathcal{W}^R_{g,t} \;=\; \sigma \sum_{s\geq 0}\Big(\mathrm{d}\mathbb{E}^{\theta}_{g,t+s}[\hat{\pi}_{t+s+1}] - \hat{\pi}_{t+s+1}\Big) \;=\; \sigma \sum_{s\geq 0}\Big[\underbrace{\Lambda^{\pi}_g \hat{\pi}_{t+s} - \hat{\pi}_{t+s+1}}_{\textit{Diagnostic error}} + \underbrace{\Lambda^{\chi}_g \hat{\chi}_{g,t+s}}_{\textit{Credibility error}}\Big]. \tag{44}$$

Denote the diagnostic error by $\mathcal{A}_{g,t}$ and the credibility error by $(1-\nu)\mathcal{B}_{g,t}$,

$$\mathcal{A}_{g,t} \;\equiv\; \sigma \sum_{s\geq 0}\big(\Lambda^{\pi}_g \hat{\pi}_{t+s} - \hat{\pi}_{t+s+1}\big), \qquad \mathcal{B}_{g,t} \;\equiv\; -\sigma \sum_{s\geq 0}(1-\rho_g)\,\Delta_g\, X_{g,t+s}, \qquad X_{g,t} \;\equiv\; \hat{\chi}_{g,t} - \theta\rho_g \hat{\chi}_{g,t-1}, \tag{45}$$

so that $\mathcal{W}^R_{g,t} = \mathcal{A}_{g,t} + (1-\nu)\mathcal{B}_{g,t}$, where $\mathcal{A}_{g,t}$ does not depend on the intensity ν. Furthermore, the ***cohort-g welfare loss*** *defined in* (42) *is, up to third-order terms, the utility the cohort loses along its chosen path relative to its rational path, measured in units of A_g, and it is the sum of the squared diagnostic error, the cross product of the two errors and the squared credibility error,*

$$L^R_{g,t} \;=\; \frac{U_g\big(c^R_{g,t}\big) - U_g\big(c_{g,t}\big)}{A_g} \;=\; \tfrac{1}{2}\mathcal{A}^2_{g,t} \;+\; (1-\nu)\,\mathcal{A}_{g,t}\mathcal{B}_{g,t} \;+\; \tfrac{1}{2}(1-\nu)^2\mathcal{B}^2_{g,t}, \tag{46}$$

while the ***aggregate welfare loss*** *of* (42) *is the cross-cohort dispersion of the cohort paths plus the squared cumulated forecast error of the consensus, Δ^R_t,*

$$\mathcal{L}^R_t \;=\; \tfrac{1}{2}\,\mathrm{Var}_{\omega}\big(dc_{g,t}\big) \;+\; \tfrac{1}{2}\big(\Delta^R_t\big)^2, \qquad \Delta^R_t \;\equiv\; \overline{dc}_t - dc^R_t \;=\; \sum_g \omega_g\, \mathcal{W}^R_{g,t}. \tag{47}$$

PROOF. See Appendix A, subsection A.7. □

Proposition 3 states that both the cohort and the aggregate welfare losses depend only on the equilibrium paths $(\hat{\pi}_t, \hat{\chi}_{g,t})$, and that the wedge is the sum of two forecast errors. First, the *diagnostic error* is the gap between the cohort's inflation forecast with the credibility channel turned off and realised inflation. It is pinned down by Λ^{π}_g, which ultimately depends on the cohort's lifetime inflation

persistence ρ_g and on θ. With under-reaction ($\theta < 0$) this error is negative for all cohorts. Thus, they expect too little inflation, perceive a real rate that is too high, and postpone consumption relative to their rational path. Importantly, this channel is not affected by how the central bank decides to communicate with the public.

Second, the *credibility error* is the part of the forecast that modulates toward which anchor the cohort inclines, $\Lambda^{\chi}_{g}\hat{\chi}_{g,t}$. This channel is positive for the cohorts that have lived with inflation above the target, because they expect more inflation, and it is the only part of the consumption wedge that the central bank can move (even eliminate) with its communication via ν. Proposition 3 also shows the channels through which the communication of the central bank may shape the cohort loss. First, the contribution of the diagnostic error, $\mathcal{A}^2_{g,t}$. Second, the two terms that the central bank can eliminate with full-target communication, namely the contribution of the credibility error, $\mathcal{B}^2_{g,t}$, and role of the cross product errors, $\mathcal{A}_{g,t}\mathcal{B}_{g,t}$.

Proposition 4 shows the consumption wedges of the HANK model. An important difference in this model is that the consumption path of cohort g additionally responds to the realised real rate through its asset position (the URE channel) and to income and taxes. However, these channels are the same under distorted beliefs and under the rational path, because both sequences face the same realised prices and income. This implies that they cancel when calculating the consumption wedge, leading the latter to depend only on the forecast error, discounted by the cohort's own consumption response.

PROPOSITION 4 (Consumption Wedges and Aggregate Welfare in HANK). *Let* $\mathrm{d}\mathbf{C}_g$ *be the cohort-g consumption path generated by* (36)*, with the perceived-rate path in* (37)*. Define the cohort-g per-capita consumption path as* $\mathrm{d}c_{g,t} \equiv \mathrm{d}C_{g,t}/C_g$*, the Jacobians* $\mathcal{J}^{c,X}_g \equiv \mathcal{J}^{C,X}_g/C_g$*, and* $\mathcal{J}^{c,\pi}_g \equiv -\mathcal{J}^{c,r^{\text{per}}}_g$*, the consumption response to the cohort's subjective inflation forecast* $\mathrm{d}\mathcal{E}_g \equiv \Lambda^{\pi}_g \mathrm{d}\boldsymbol{\pi} + \Lambda^{\chi}_g \mathrm{d}\boldsymbol{\chi}_g$*, with* Λ^{π}_g *and* Λ^{χ}_g *the stacked operators of* (43)*. Let* $\mathrm{d}\mathbf{c}^R_g$ *be the* ***rational path*** *of cohort g: its per-capita response in the same economy when it forecasts the realised inflation path with perfect foresight,* $\mathrm{d}\mathcal{E}^R_g = \mathbb{F}\,\mathrm{d}\boldsymbol{\pi}$ *with* $(\mathbb{F}\mathbf{x})_t \equiv x_{t+1}$ *the lead operator, at the same realised paths* $(\mathrm{d}\boldsymbol{\pi}, \mathrm{d}\mathbf{r}, \mathrm{d}\mathbf{X})$*,* $X \in \{i, Y, \tau\}$*. Then, the cohort consumption wedge* $\mathcal{W}^{H,R}_{g,t} \equiv \mathrm{d}c_{g,t} - \mathrm{d}c^R_{g,t}$ *is the cohort's cumulated forecast error passed through its own consumption response and splits exactly into a* ***diagnostic error*** *and a* ***credibility error****:*

$$\begin{aligned}\mathcal{W}^{H,R}_{g,t} &= \Big[\mathcal{J}^{c,\pi}_g\big(\mathrm{d}\mathcal{E}_g - \mathbb{F}\,\mathrm{d}\boldsymbol{\pi}\big)\Big]_t \\ &= \underbrace{\Big[\mathcal{J}^{c,\pi}_g\big(\Lambda^{\pi}_g - \mathbb{F}\big)\mathrm{d}\boldsymbol{\pi}\Big]_t}_{\textit{Diagnostic error}} + \underbrace{\Big[\mathcal{J}^{c,\pi}_g \Lambda^{\chi}_g\,\mathrm{d}\boldsymbol{\chi}_g\Big]_t}_{\textit{Credibility error}}.\end{aligned} \tag{48}$$

the balance-sheet (URE) and $\{i, Y, \tau\}$ channels cancel because both paths face the same realised prices and income. Moreover, with the cohort and the aggregate welfare losses of (42) *evaluated at $\mathcal{W}^{H,R}_{g,t}$, the aggregate loss is*

$$\mathcal{L}^{R}_t = \tfrac{1}{2}\,\mathrm{Var}_{\omega}\big(\mathcal{W}^{H,R}_{g,t}\big) + \tfrac{1}{2}\big(\Delta^{H,R}_t\big)^2, \qquad \Delta^{H,R}_t \equiv \sum_g \omega_g \mathcal{W}^{H,R}_{g,t} = \overline{\mathrm{d}c}_t - \overline{\mathrm{d}c^R}_t,$$

the dispersion of the cohort forecast errors plus the squared consensus error.

PROOF. See Appendix A, subsection A.8. □

Proposition 4 shows that the HANK wedge has the same two parts as the RANK wedge, a diagnostic error and a credibility error. However, the HANK model changes the wedge in one way. Since the diagnostic and credibility errors are scaled by the cohort's consumption Jacobian $\mathcal{J}^{c,\pi}_g$, they move consumption by more in a high-MPC cohort than in a low-MPC one, and the consumption response is spread over time rather than taken all at once, since households close to the borrowing constraint cannot adjust their consumption immediately. Balance sheets and income still matter through the rational path $dc^R_{g,t}$. Since younger cohorts, which hold the smallest asset positions, depend most on labour income and have the highest MPC, they cut consumption the most when income falls after the shock. However, this is the incidence of the shock, not a welfare loss from beliefs.

FIGURE 5. Cohort Consumption Wedges After a +1pp Cost-Push

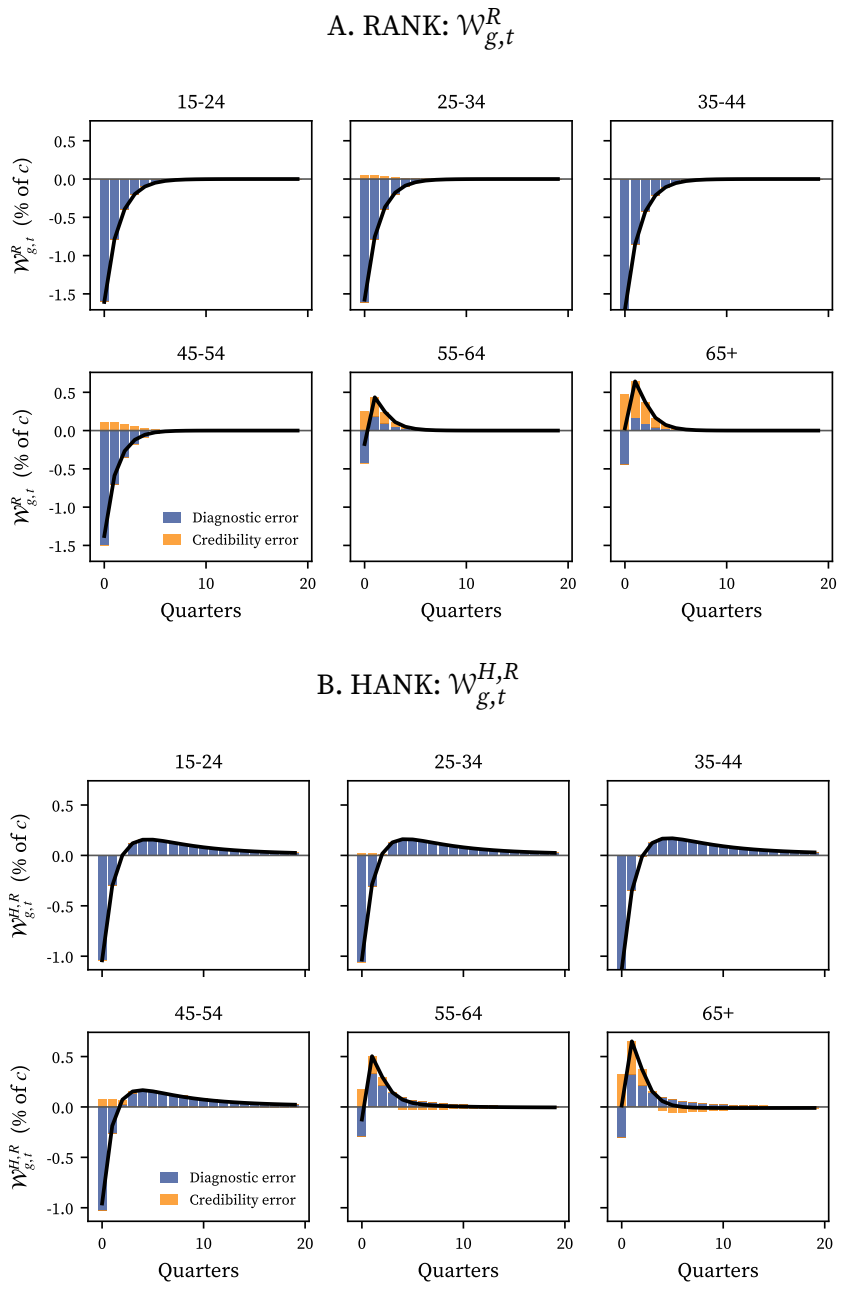


Figure 5 shows the cohort wedges after the cost-push shock, and both models confirm the same story. For the younger cohorts (under 55) the diagnostic error of the wedge is negative, implying that they under-react to the shock, expect too little inflation and under-consume on impact relative to their rational path. Their credibility error is negligible, because their experienced inflation is close to the target. For the older cohorts (55 or above), the credibility error is positive and

large enough to reverse the sign of the wedge after impact. Since they have lived with inflation consistently above the target, they expect more, and over-consume. The HANK model has smaller wedges, because only unconstrained households substitute intertemporally, while the hand-to-mouth cannot.

5.2. Communication Design

This section asks which intensity of target-communication the central bank should choose and, counterintuitively, the answer is not full intensity. Communicating the target removes the part of each cohort's forecast that follows its own inflation experience, but raises the consensus error. After an inflationary shock, full target communication incorrectly anchors cohorts that would have expected a higher inflation path, closer to the realised one. The optimal communication intensity balances these two effects, and the result is different between the RANK and the HANK model. This section also shows how the gain from communication is distributed across cohorts, how communication interacts with traditional monetary policy, and the costs of delayed target communication.

5.2.1. Optimal Target Communication

Propositions 3 and 4 show that only the credibility error reacts to ν and it is higher for the cohorts with a high inflationary experience, whose forecasts lean toward their own history when a cost-push shock hits. Therefore, for these cohorts the credibility error is positive and the diagnostic error is negative due to under-reaction, so these two effects partly cancel each other out. This generates the paradox that their inflation history pushes them to over-react against the under-reaction of diagnostic expectations. Therefore, removing the credibility error with communication could raise the error of these cohorts, whose inflation experience paradoxically helps them to over-react, and then increase the error of the consensus. Proposition 5 makes this trade-off explicit.

PROPOSITION 5 (Optimal Target Communication). *Consider a cost-push shock with realised, already observed, equilibrium paths $\{\hat{\pi}_t, \hat{\chi}_{g,t}\}_{t\geq 0}$ of inflation and cohort credibility.*

a. RANK Model. *Along the realised paths the cohort-g wedge of Proposition 3 is linear in the intensity, $\mathcal{W}^R_{g,t}(\nu) = \mathcal{A}_{g,t} + (1-\nu)\,\mathcal{B}_{g,t}$, with $\mathcal{A}_{g,t}$ and $\mathcal{B}_{g,t}$ the diagnostic and credibility errors of* (45)*: $\mathcal{A}_{g,t}$ does not depend on ν and the credibility error*

$(1-\nu)\mathcal{B}_{g,t}$ *scales with the distance from full intensity. Let the cumulated losses be* $L_g^R(\nu) = \frac{1}{2}\sum_{t\geq 0}\big(\mathcal{W}_{g,t}^R(\nu)\big)^2$ *and* $\mathcal{L}^R(\nu) = \sum_g \omega_g L_g^R(\nu)$*. Then the intensity cohort g would choose and the aggregate optimum are*

$$\nu_g^\star \equiv \arg\min_{\nu\in\mathbb{R}} L_g^R(\nu) = 1 + \frac{\sum_{t\geq 0}\mathcal{A}_{g,t}\mathcal{B}_{g,t}}{\sum_{t\geq 0}\mathcal{B}_{g,t}^2},$$
$$\nu_{\text{agg}}^\star \equiv \arg\min_{\nu\in\mathbb{R}} \mathcal{L}^R(\nu) = 1 + \frac{\sum_g \omega_g \sum_{t\geq 0}\mathcal{A}_{g,t}\mathcal{B}_{g,t}}{\sum_g \omega_g \sum_{t\geq 0}\mathcal{B}_{g,t}^2},$$

whenever the denominators are positive. Full intensity, $\nu^\star = 1$*, is optimal only when the two error paths are orthogonal,* $\sum_{t\geq 0}\mathcal{A}_{g,t}\mathcal{B}_{g,t} = 0$*. The optimum is interior,* $\nu^\star < 1$*, when the cross product is negative,* $\sum_{t\geq 0}\mathcal{A}_{g,t}\mathcal{B}_{g,t} < 0$*, that is, when the two errors have opposite signs in the quarters where they are large, so that the credibility error partly offsets the diagnostic one. The ratio* $\sum_{t\geq 0}\mathcal{A}_{g,t}\mathcal{B}_{g,t}/\sum_{t\geq 0}\mathcal{B}_{g,t}^2$ *is the slope of a regression of* $-\mathcal{A}_{g,t}$ *on* $\mathcal{B}_{g,t}$ *through the origin. The cohort optima do not coincide in general.*

b. HANK Model. *Along the realised paths the cohort-g wedge of Proposition 4 is linear in the intensity,* $\mathcal{W}_{g,t}^{H,R}(\nu) = \mathcal{A}_{g,t}^H + (1-\nu)\mathcal{B}_{g,t}^H$*, with*

$$\mathcal{A}_{g,t}^H \equiv \Big[\mathcal{J}_g^{c,\pi}\big(\Lambda_g^\pi - \mathbb{F}\big)\mathrm{d}\boldsymbol{\pi}\Big]_t,$$
$$\mathcal{B}_{g,t}^H \equiv -(1-\rho_g)\,\Delta_g\left[\mathcal{J}_g^{c,\pi}\mathbf{X}_g\right]_t, \qquad \mathbf{X}_g \equiv (\mathbb{I} - \theta\rho_g L)\,\mathrm{d}\boldsymbol{\chi}_g.$$

With the same cumulated losses, $\nu_g^\star$ *and* $\nu_{\text{agg}}^\star$ *take the same form with* $(\mathcal{A}^H, \mathcal{B}^H)$ *in place of* $(\mathcal{A}, \mathcal{B})$*, as in the RANK model.*

PROOF. See Appendix A, subsection A.9. □

Proposition 5 makes precise what target communication can and cannot do. In both models the wedge is linear in ν. Then, the diagnostic error $\mathcal{A}_{g,t}$ is fixed and the credibility error $(1-\nu)\mathcal{B}_{g,t}$ shrinks as the communication intensity increases. Full intensity removes the credibility error of every cohort, but it is optimal only when the two errors are orthogonal, as for the young cohorts, whose experienced inflation is close to the target and whose credibility error is negligible. When the errors are not orthogonal, as for the older cohorts with higher experienced inflation, the credibility error partly offsets the diagnostic error and this calls for less intensive communication. This is in line with the consumption wedges previously mentioned, which require cohorts to lean more on experienced inflation to keep

them closer to the realised path. Since the cross product of the errors is cohort specific, the cohort optima in general do not coincide.

The aggregate optimum should weight them and capture the importance each cohort has in pinning down the policy the CB should choose. In the HANK model both errors are scaled by $\mathcal{J}_g^{c,\pi}$, so heterogeneous MPCs change the size of each error and the weight of each cohort in the aggregate optimum, but the expression for the optimal intensity remains the same. Hence, the CB faces limitations. First, it cannot reach the diagnostic channel, since ν only affects κ_g but never ρ_g or θ. This implies that the dispersion of Λ_g^{π} in the wedges is independent of how the CB communicates. Additionally, in the HANK model the CB cannot affect the cohorts' asset positions and MPCs, which provides additional support to the fact that the the credibility channel is the unique it may reach. The discussion on the optimality of target communication connects with its costs and benefits after an inflationary shock. Consider the benefit first. After the shock, cohorts disagree because each one inclines towards its own inflation experience. Thus, anchoring them to the inflation target makes their beliefs more similar, so the dispersion of the consumption wedges falls. This dispersion is the first term of the aggregate loss in (47), $\frac{1}{2}\text{Var}_\omega(\mathcal{W}_{g,t}^R)$, that captures how far each cohort's consumption departs from the average because its forecast error differs from the average forecast error.

However, the part of the forecast that the central bank removes is not noise for the consensus. After the impact, the older cohorts lean on their high experienced inflation $\bar{\pi}_g$ through the credibility channel and over-predict inflation. For them this is an idiosyncratic error and removing it lowers their loss. However, for the consensus it was the only part of the average forecast that followed the inflation that was actually happening. When it is removed, no cohort tracks realised inflation any more, and the consensus under-predicts its path for longer. This cost is common to all cohorts, while the benefit of removing the bias is concentrated in the older ones. This situation represents a cost of communicating, captured by the consensus error in (47), Δ_t^R, which increases. The cost differs across models, and the reason is closely related to the size of the consumption wedges, which were larger in the RANK than in the HANK. In the RANK model the effect of the consensus error is more prolonged, since it is incorporated along the entire consumption path, because the permanent-income household responds to the whole path of expected real rates. However, in the HANK model many households are constrained or substitute little over time, so their consumption responds mainly to

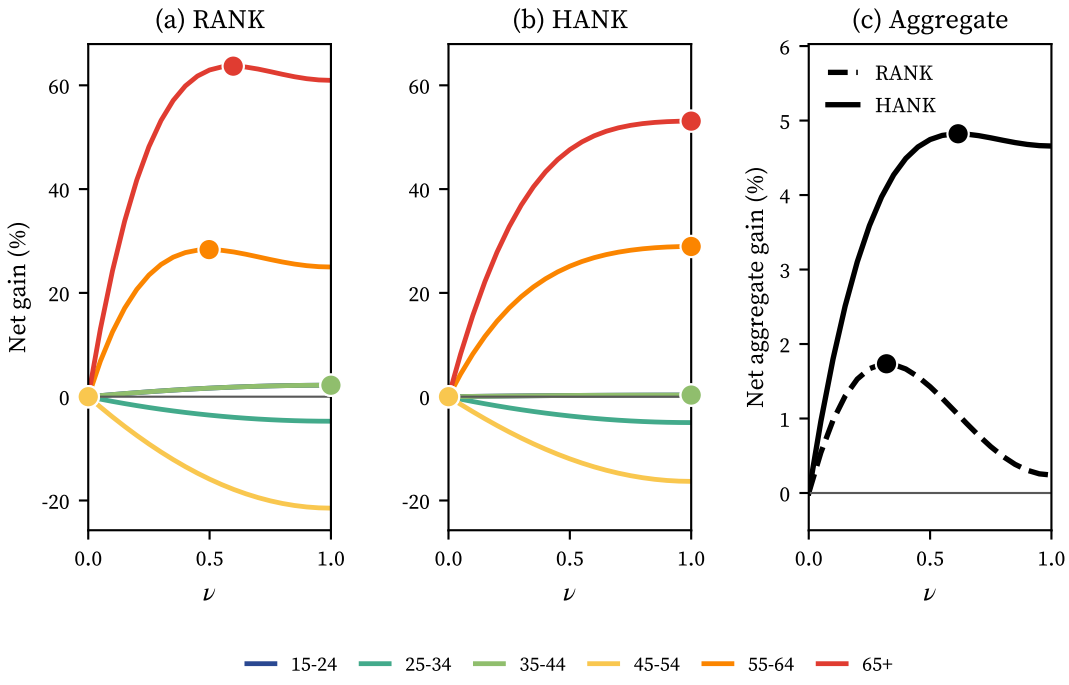
the current error. This makes the effect of the consensus error be absorbed quickly, since even if households wanted to spread it along their future consumption path, they could not, because many of them are liquidity constrained.

To measure the cohort-g net gain from the intensity of communication, this paper uses the following metric based on the rational path:

$$\text{Net Gain}_g(\nu) \equiv 100\left(\frac{L_g^R(0) - L_g^R(\nu)}{L_g^R(0)}\right), \tag{49}$$

that is, how much the loss of the cohort improves with the intensity of communication ν relative to the case of no communication at all. Panels a) and b) of Figure 6 shows that the two cohorts over 55 gain the most since communication makes their credibility error on the wedges (the largest) disappear, and the fall in their dispersion is larger than the consensus error they take on. On the other hand, younger cohorts (with the exception of the 45–54, which actually loses) gain almost nothing, since they had no credibility error to remove, and the consensus error they take on is similar to the dispersion they lose due to communication.

FIGURE 6. **Net Gain from Target Communication**



Note: Dots show the optimal ν.

As anticipated, the optimal communication each cohort requires is different, and it changes with the model under consideration. The most significant change occurs in the old cohorts, which in the RANK choose an interior solution for ν, since on impact their credibility error is positive and offsets their negative diagnostic error, so keeping part of the credibility channel is actually optimal. However, in the HANK they choose a corner solution with $\nu = 1$, since there both errors push consumption in the same direction. The reason entirely comes from income heterogeneity. The fall in labour income after the shock strongly hits high-MPC households, which cannot smooth consumption, so the contraction in demand is persistent, affecting the inflation path, which returns to target faster than the shock alone would imply.

Thus, the old cohorts over-predict inflation for several periods after impact, making their diagnostic error turn positive precisely when their credibility error is also positive. In contrast with the RANK model, where partially keeping the credibility channel turns out to be optimal, what the central bank should do is to remove the credibility error entirely. Panel (c) aggregates these cohort gains with the population weights. The two old cohorts, which carry the largest credibility error and more than a third of the population, drive the result, and indeed the aggregate reflects this by leaning, for the same reasons, towards the optimum these cohorts demand.

5.2.2. Communication, Monetary Policy and Timing

Once the optimal target communication is characterised, this paper examines two additional topics. First, how target communication interacts with the traditional interest-rate adjustments, analysed by Goy et al. (2022) for forward guidance under imperfect credibility and empirically by Coibion et al. (2022). Second, once the economy is hit by an inflationary shock, what the associated costs of delayed communication are, in the timing sense of Bomfim and Rudebusch (2000) for disinflation under imperfect credibility.

Addressing the first inquiry, Figure 7 compares how target communication interacts with the inflation coefficient of the Taylor rule, ϕ and its effects on the aggregate loss. This figure shows the norm of the consumption wedges, which is defined by:

$$\left\|\mathcal{W}^R\right\|_\omega = \sqrt{\sum_g \omega_g \sum_t \left(\mathcal{W}^R_{g,t}\right)^2}. \tag{50}$$

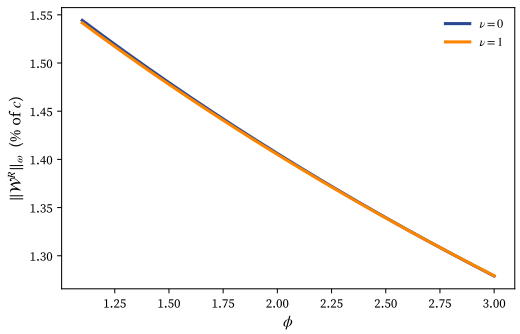
and measures the cohort-weighted distance to the rational path after an inflationary shock.

In the RANK model, panel A, a more aggressive rule lowers the loss, regardless of the communication intensity, since restrictive monetary policy reduces the dispersion of the wedges and the consensus error in the same proportion. While communication removes the credibility error, the consensus error rises by the same amount at every ϕ, thus there is no substantial benefit from full intensive communication

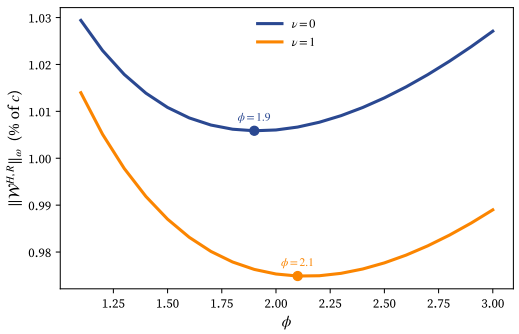
The story is different in the HANK model, as panel B shows. As in RANK, a more aggressive monetary policy rule reduces inflation on impact and lowers the dispersion of the wedges. However, since the fall in labour income of the high-MPC households reduces aggregate demand even more, inflation falls below the target sooner and more drastically. Thus, cohorts that forecast with their own inflation persistence keep expecting positive inflation when it has already turned negative. Contrary to RANK, communication is helpful since experienced inflation is precisely what makes old cohorts expect higher inflation when it is actually decreasing. Thus, removing this over-prediction with intense communication is what makes an aggressive rule appealing. Importantly, full target communication and a moderately aggressive ($\phi = 2.1$) monetary policy deliver the lowest welfare loss, which is evidence that both policies may be used as complements by the central bank.

FIGURE 7. Target Communication and Traditional Monetary Policy

A. RANK



B. HANK



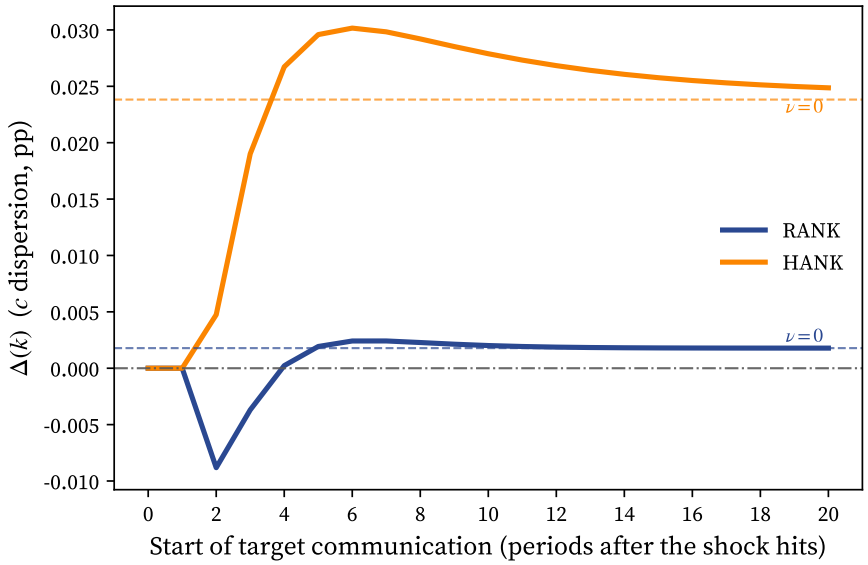
Regarding the timing question, this paper considers the aggregate loss when full target communication ($\nu = 1$) starts k periods after the shock, against the case in which it starts on impact ($t = 0$). Let $\nu^{(k)}$ be the intensity path that adopts full target communication k periods after the shock hits the economy. The cost of the delay is

$$\Delta(k) \equiv \|\mathcal{W}^R\|_\omega(k) - \|\mathcal{W}^R\|_\omega(0), \tag{51}$$

where $\|\mathcal{W}^R\|_\omega(k)$ is the norm of (50) on the equilibrium path under $\nu^{(k)}$, so that $\Delta(k)$ is the extra loss of starting at k instead of on impact.

Figure 8 plots the cost of delayed communication and shows that in the RANK model a short delay is better than communicating on impact. Since the forecasts of the older cohorts are the only part of the consensus that follows realised inflation, allowing that part of their inflation experience to be incorporated into the consensus, although on impact is costly because of the dispersion it keeps, turns out to be beneficial, as it lowers the consensus error over time. Once this period has passed, waiting to communicate increases the cost, given the potential credibility erosion. On the contrary, in the HANK model starting on impact is best. Even when early communication raises the consensus error, it removes the previously discussed inflation over-prediction of the older cohorts, which dominates. Thus, reacting early prevents credibility from eroding, so the target has to be communicated once the shock hits, not once its effects on expectations are visible.

FIGURE 8. Timing of Target Communication



6. Conclusion

Central banks respond to inflationary episodes by constantly repeating that they will do everything in their power to bring inflation back to the target. While this message seems to reach the most sophisticated individuals, households apparently do not notice these announcements when forming their expectations. This paper builds a model of inflation expectations with endogenous credibility that is consistent with the under-reaction of households by incorporating DE as a fundamental part of how households process news, and then applies this mechanism to the New Keynesian model, both in its simple version and in its heterogeneous-agent version.

Three lessons emerge from this paper. The first is related to who pays more in terms of welfare after an inflationary shock. In both models the cost of distorted beliefs falls on the cohorts whose memories diverge the most from the target, namely the old cohorts. The second lesson is that intensively communicating the target is not always optimal. Communication removes the credibility error and lowers the dispersion of beliefs across cohorts, but the forecasts it removes were the only part of the consensus that followed realised inflation, so the error of the consensus rises. The optimal intensity balances both effects and is higher in the HANK mode, where the old cohorts over-estimate inflation for longer and the

gain from anchoring them dominates. The third lesson is about the connection between target-based communication and traditional monetary policy. With a representative agent, a more aggressive interest rate rule lowers the loss regardless of communication. However, with heterogeneous agents, the stronger demand contraction of constrained households brings inflation below the target sooner, causing old cohorts to over-predict inflation, being communication an effective tool to prevent it. Consequently, the target has to be communicated when the shock hits and not once its effects on expectations are visible on credibility.

This paper leaves some open questions. The communication intensity is a choice of the CB, but its reception by households is taken as given. Studying theoretically how the central bank can reach households with its communication is a research area that would complement this paper. Another pending area comes from relaxing the assumption that inflation experiences are fixed over time. Allowing recent inflation to be incorporated into experienced inflation presents a methodological challenge, but it would evidently add realism to this analysis. Finally, extending the mechanism to the interaction between target communication and forward guidance on the policy rate would offer a wider spectrum of communication policies that the CB can use under distorted expectations.

Online Appendix

The Online Appendix is available **here.**

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